The Bluffs North Expansion (Manistee Lake)

AI-generated review: This review was generated by AI, so some of the language may sound blunt or overly direct, and some of the findings may simply be wrong -- please don't take any of it personally, and verify anything that matters before acting on it.

As of 2026-09-28 · Basis: Spicer Group D-6631, 03A Plan Set, 16ft BOC Alternate, REV 3, sealed, IFB 8/20/26 (49 sheets)

PROGRAM

155 New sites	**63** MH double-wide	**81** MH single-wide
11 RV super sites	**26.1** Net lot acres	**32–33 ac** Gross acres (est.)

The Ideas doc counts 144 home sites + 11 RV super sites (155 total) from the sheet 06 parcel tables, and flags that the electrical sheets only show symbols for about 101 home services and 7 RV pedestals -- a real discrepancy Marc flagged for Spicer to confirm, not yet resolved.

COST ESTIMATES (4)



AACE Class 4 estimate [class4]

$8,971,433 Total project cost	**$57,880** Per site	**155** Sites assumed	**$7.18M–$11.66M** Class range

Regional factor: 0.92 to 0.95

Component	Amount
Direct hard cost	$6,503,858
Hard contingency 12%	$780,463
Soft costs incl. financing	$1,533,738
Soft contingency 10%	$153,374

Phase 1: $5,390,000 (60 sites) — placeholder: 60 sites and 60% of the linear network

Cost and VE base case (Class 3, per the full memo) [class3_ve]

$8,505,400	$54,900	155	$6.80M–$11.15M
Total project cost	Per site	Sites assumed	Class range

Regional factor: 0.90 (Northern Lower Michigan)

Component	Amount
Direct hard cost	$5,772,100
General conditions 6% + bonds/insurance 1.5%	$432,900
Hard-cost contingency 12%	$744,600
Soft costs (22% of hard)	$1,555,800

Phase 1: $3,900,000 (60 sites) — ~$65k/site; front-loads the Oregon trunk sewer to EX MH #2, water ties, and Oregon/California/Kentucky roads

Recommended alternative: 165 sites, $7,780,000 ($47,200/site, $52,600/site like-for-like). 165 sites (154 MH + 11 RV), driven mainly by B4 water/sewer submetering (+$75k cap/+$66k NOI), A9 driveways/shed slabs shifted into the home-sale package (-$672k developer capital), B1 density +10 sites (+$175k/+$41k NOI), B2 RV/boat storage on deep-lot remnant (+$119k/+$28k NOI), and a spec-rationalization bundle (-$212k). Stabilized NOI +$135k/yr (directional), yield on cost 7.2% -> 9.5%.

Class 4 Estimate, Rev 1 -- Long-Term Hold (as-drawn 155 sites, durability specs applied)

[class4_rev1_long_term_hold]

$11,866,642	$76,559	$9.49M–$15.43M
Total project cost	Per site	Class range

Regional factor: 0.92 to 0.95

Phase 1: $7,130,000 (60 sites)

Cost & VE Memo, Rev 2 -- Recommended Long-Term-Hold Plan (165 sites, durability + revenue-positive ideas kept)

[cost_ve_rev2_long_term_hold]

$9,600,000	$58,200
Total project cost	Per site

Regional factor: 0.90 (Northern Lower Michigan)

Phase 1: $4,800,000 (60 sites)

REAL BID VS. THIS PAGE'S ESTIMATES

Awaiting the real contractor bid (expected 2026-10-07) — Andrew Chime (Grex Dev), email 2026-10-02: bids due Oct 6 noon ET, comparison delivered ~Oct 7, reviewed with Jay/Marc Oct 8

Placeholder built 2026-10-05, ready to fill in the moment Grex Dev's real comparison lands. This is the first time any of this pipeline's three reviewed projects (Bluffs, Madison, Madison Clubhouse) will have a REAL contractor bid to check an AI cost estimate against -- treat filling this in as a real deadline, not a someday task.

Estimate	Total
Class 4 estimate (base, as-drawn)	$8,971,433
Cost & VE Memo base case (Class 3)	$8,505,400
Class 4 Rev 1 -- Long-Term Hold	$11,866,642
Cost & VE Memo Rev 2 -- recommended Long-Term-Hold plan	$9,600,000

THE BIG PICTURE

1. What will this cost?

PRIMARY ANALYSIS (CLAUDE)

Expect roughly **$8.5M–$9.0M all-in, or about $55K–$58K per site** — two independent estimates on the same plan set came in at $8.51M (Class 3, Cost & VE Memo) and $8.97M (Class 4 Estimate), a $466K gap. Neither should be treated as final because both flag the same two missing numbers: what the sewer/water system will charge for 155 new connections, and what Consumers Energy will bill for the electric line extension — together those two unknowns could swing the total by $0.5M–$1.2M in either direction. Phase 1 (roughly 60 sites) runs $3.9M–$5.4M, front-loaded because the trunk sewer, water ties, and main roads have to go in first.

INDEPENDENT SECOND OPINION (CLAUDE, ZERO EXPOSURE TO THE ABOVE)

My independent figure: $14.2M all-in ($91,600/site), with a hard cost base of $10.26M before contingency.

That's meaningfully higher than the published $8.5M–$9.0M range. Here's where the gap comes from — and I want to be direct about what's assumption versus what's plan evidence.

What drives my higher number:

- **Amenity buildings ($1.64M).** I included a bathhouse, clubhouse, office, and maintenance building because a 155-site resort cannot get a Manistee County campground license without at minimum a bathhouse and dump station. If the published estimate excludes vertical structures entirely (a reasonable scope choice if those are a separate contract), that single line item explains most of the $5M–$6M gap between our totals. I don't think the buildings can be ignored in a true all-in cost, but I acknowledge the published estimate may be scoping only the civil infrastructure package.
- **Contingency (20% vs. whatever the published estimate used).** I set contingency at 20% of hard cost ($2.05M) specifically because this plan set has six critical unresolved issues, two missing permit items, and multiple unreadable sheets. That's not normal contingency territory — it's a plan set where the sanitary system may not drain, where no stormwater detention exists anywhere in the documents, and where the geotechnical basis is entirely absent. A 10–15% contingency on a plan set this incomplete understates the risk.
- **Error register allowances baked in.** I included $200K for a potential lift station (E-01), $250K for a detention basin EGLE will almost certainly require (E-05), and used 200A electrical service for MH lots per NEC 550.32 rather than the

100A shown on D-6631-49. These aren't gold-plating — they're filling gaps the plan set left open.

What I agree with in the published estimate: The utility tap/connection fee uncertainty is real and flagged in both analyses. My soft cost line includes $550K for water and sewer connection fees — a wide-ranging estimate that I flagged as the single most variable soft cost item.

Where we agree on confidence: Both analyses acknowledge this is an AACE Class 4 exercise at ±30–50%. At the low end of my range (removing amenity buildings, removing lift station allowance, confirming no detention required): my estimate drops to roughly $9.8M. At the high end of the published range with those items added back: you get close to $10.5M–$11M. The overlap exists — we're not describing different projects, we're making different scope inclusions.

The single biggest unknown that neither estimate can resolve: The MH pad type. D-6631-06 (Parcel Types sheet) is a visual-only page that I could not read. If it shows full concrete mat pads for the ~75–80 MH sites, that's $3,000–$6,000/site in concrete — roughly $450K–$480K in aggregate. If it shows a standard runner system, that drops to under $150K. That one unread sheet swings the estimate by $300K–$600K. **This is not in the published material and it is the highest-priority verification item before trusting any cost number from either analysis.**

A finding not in the published material: The transformer procurement timeline is a schedule risk that functions as a cost risk. Pad-mount distribution transformers are on 40–80 week allocation nationally as of mid-2026. If the owner doesn't order transformers before mobilization — potentially before bid award — the electrical scope stalls and every downstream trade (pedestals, secondary distribution, panel sets) sits idle on a mobilized site. That's a real carrying cost on a $14M project. Neither analysis I've seen treats transformer lead time as a budget line, but it should be: 2–3 weeks of idle site overhead on a fully mobilized crew runs $40K–$80K.

INDEPENDENT SECOND OPINION (CROSS-MODEL, GOOGLE GEMINI)

I cannot provide a definitive total project cost, as the exact figures cannot be reliably estimated from the text provided. To estimate this accurately, we need a complete tabulation of cut/fill earthwork volumes, individual utility lateral lengths for each pad, and linear footages of road paving, none of which are fully quantified in the provided plan set text.

Using naive area-times-depth math for civil works or pricing only the utility trunk lines while ignoring the dozens of individual site services systematically underestimates the true cost of development.

However, based on the scope, we can establish a conceptual baseline. * **OBSERVED**: The plans include a mix of Manufactured Home (MH) and RV super sites across roughly 73+ new primary parcels, plus upgrades (concrete pads) to several existing sites (Lots 132-135, 136-138, 148-155) (Sheet 06). * **OBSERVED**: Trunk infrastructure includes approx. 7,600 LF of 6" C900 PVC water main (summed from Sheets 32-43), extensive 8" PVC sanitary main (Sheets 22-30), and 12"-36" HDPE storm sewer (Sheets 12-20). * **ASSUMED**: For a premium MH/RV community expansion in Michigan utilizing public utilities and paved roads, fully burdened development costs typically range from $35,000 to $55,000 per pad (excluding land acquisition). * **UNKNOWN**: Total linear footage of 4" or 6" individual sanitary/water laterals. Total cubic yards of earthwork cut/fill. Primary electrical trenching costs (Sheet 49 notes conduit and conductors from the utility are "VIF" - Verify In Field).

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2. What can we do to reduce it?

PRIMARY ANALYSIS (CLAUDE)

The single best-supported move is **shifting driveways and shed slabs into the home-sale package** — that alone saves $672K in developer capital by pricing it into each home instead of carrying it in the construction budget. Close behind: the Cost & VE Memo's recommended 165-site plan (adding 10 sites with tighter lot frontages) drops total cost to $7.78M and raises yield on cost from 7.2% to 9.5%, driven mainly by the density gain plus submetering revenue — though it requires a replat and EGLE resubmittal before bid award.

INDEPENDENT SECOND OPINION (CLAUDE, ZERO EXPOSURE TO THE ABOVE)

My independent finding: the five highest-confidence savings are worth $812K–$1.28M in hard cost reduction, and three of them require zero plan amendments.

Here's what my review turned up, ranked by confidence and net impact:

1. MH runner pads vs. full concrete mat — up to $603K savings (verify D-6631-06 first) If the approved plans show full concrete mat for MH sites, switching to the standard HUD-compliant runner system (two 16"-wide reinforced concrete runners per site, 6" depth, with compacted aggregate fill between) saves $500K–$603K in concrete costs and is fully compliant with 24 CFR Part 3285. This is the largest single VE lever on the project. But — and this matters — if D-6631-06 already shows a runner system, the savings is $0. You have to open that sheet before this number means anything. I flagged this as the highest-priority verification item in my review. **This is not in the published material.**

2. Chip seal on secondary streets — $226K–$265K savings, high confidence Replace full-depth HMA on Kentucky, Minnesota, Montana, Pennsylvania, and Washington Avenues (the five cross-streets serving primarily RV sites) with chip seal. Retain full-depth asphalt on California Drive and Tennessee Avenue — those collector streets will take MH transport trucks, fire apparatus, and RV coach traffic for decades and need the structural section. The secondary streets don't. Chip seal is appropriate for RV-only roads, costs $18–$22/SY installed vs. $45/SY for HMA, and requires recoat every 5–7 years at $4–$6/SY — still cheaper at a 20-year life-cycle horizon. No plan amendment, no agency resubmittal.

3. Joint trenching — water main + electrical conduit in common trench: $86K–$213K savings Michigan Act 399 requires 10 ft horizontal separation between water and sanitary mains, so those two can't share a trench. But electrical primary conduit can share a trench with the water main (12" minimum separation per most utility provider standards). Running both in one open cut instead of two eliminates one full trenching mobilization per street — that's real money at $18–$25/LF per trench on 6,100 LF of road. Confirm with the serving utility before bid; some co-ops prohibit shared trenches. If approved: no plan amendment needed beyond a joint-trench detail sheet. **Schedule benefit too — one mobilization instead of two per street, roughly 2–3 weeks compression.**

4. Investigate gravity feasibility on California Drive before accepting a lift station — $100K–$270K net savings if regrading avoids it This is a design optimization, not a substitution. The low point on California Drive is at El. 659.22 (D-6631-13, D-6631-23). If the existing sanitary sewer invert at the POE sits above that elevation, the gravity system doesn't work and a lift station adds $180K–$350K to the base contract — money not in any bid document right now. But before accepting that outcome, the EOR should check whether raising the California Drive road profile by 1–2 ft over 400–600 LF makes gravity feasible. Regrading that reach costs $30K–$80K in revised earthwork, which is far cheaper than a lift station plus its 20-year O&M liability. **This is the first RFI that should go out, before any trench is opened.**

5. Utility submetering — not a cost reduction, but a capital reallocation with a 0.4–0.9 year payback The published analysis talks about shifting driveways into the home-sale package as the top VE move. That's a reasonable approach for the MH side. My independent read identified submetering as the complementary move on the revenue side: installing smart electric meters at all 155 sites adds $31K–$78K to capital but recovers $83K–$88K/year in utility cost pass-through. At a 7.5% cap rate that's $1.1M–$1.2M in property value on an $55K–$78K investment. **The critical constraint: meter modules must be specified before pedestal orders go out — 8–12 week lead time — and Michigan PSC submetering regulations must be confirmed before you commit to the billing structure.** Retrofitting meter sockets after installation costs 3–5× as much.

One genuine disagreement with the published analysis's framing: The published material leads with shifting driveways and shed slabs into the home-sale package as the "single best-supported move." That's a legitimate capital allocation strategy, but it doesn't reduce the total project cost — it shifts who pays for the concrete from the developer's construction budget to the homebuyer's purchase price. If the goal is reducing developer capital at risk, yes, it works. If the goal is reducing total dollars spent on the project, it doesn't — the concrete still gets poured. My runner-pad finding (if D-6631-06 shows full mat) actually eliminates concrete rather than reallocating it, which is a different and potentially larger benefit. Both approaches are valid; they're answering different questions.

INDEPENDENT SECOND OPINION (CROSS-MODEL, GOOGLE GEMINI)

As a permanent owner-operator holding this community for 40-50 years, true value-engineering means minimizing the Total Cost of Ownership (TCO). Several items in these plans represent short-term savings that will create massive long-

term liabilities. To reduce our 40-50 year cost, we must reject the following "cheap" upfront designs and upgrade them immediately:

- **Reject Asphalt Integral Curbs:**
  - **OBSERVED**: Sheet 10 specifies a "7" INTEGRAL ASPHALT CURB (TYP)".
  - **The 50-Year Lens**: Asphalt curbs are a massive liability in an MH/RV park. Heavy MH delivery toters, turning RVs, and winter snowplows (Manistee, MI gets heavy snow) will obliterate asphalt curbs within 3-5 years. You will spend decades patching them, and they will never look right.
  - **Recommendation**: Spend the money now to upgrade to standard MDOT concrete roll curbs (or high-back drop curbs). The upfront cost is higher, but the 40-year maintenance cost drops to near zero.
- **Reject 3-Inch Pavement Thickness:**
  - **OBSERVED**: Sheet 10 dictates 1.5" 4EL leveling course and 1.5" 5EL wearing course (3" total HMA) over 8" 21AA aggregate.
  - **The 50-Year Lens**: 3" of asphalt is standard for light residential subdivisions, but MH/RV parks see concentrated heavy axle loads from Class A motorhomes, garbage trucks, and home deliveries. 3" will likely rut and alligator-crack prematurely.
  - **Recommendation**: Increase to 4" total HMA (2" base, 2" top). The upfront asphalt tonnage increases by 33%, but you will push your first major mill-and-overlay cycle from year 10-12 back to year 20+.
- **Rigid QA/QC on Storm Sewer Bedding:**
  - **OBSERVED**: Sheets 12-20 call for HDPE storm sewers. Sheet 21 requires MDOT 6A stone backfill under roads.
  - **The 50-Year Lens**: HDPE is cheaper than Reinforced Concrete Pipe (RCP) but relies 100% on perfect trench compaction to avoid deflecting (egg-shaping). If the contractor cheats and uses native sand without proper lifts, the pipe will fail under the road.
  - **Recommendation**: You can keep the HDPE to save money, but you must strictly enforce the 6A stone bedding (OBSERVED on Sheet 21) with an independent soils inspector on site. Alternatively, upgrade to RCP under the paved roadways.

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3. What else would you recommend?

PRIMARY ANALYSIS (CLAUDE)

Stop the clock on two things before you break ground. First, get a written capacity letter from whoever owns the sewer system downstream of existing MH #2 — if that system can't take 155 new connections, your entire gravity-sewer design is stranded and the cost swings by up to $1.2M. Second, pull up the EGLE High Risk Erosion Area GIS layer and call the Township zoning administrator to confirm whether this bluff above Manistee Lake triggers Michigan's HREA setback program — it's written for Lake Michigan shoreline, but Manistee Lake connects to it, and if the 60-year recession setback applies, your buildable envelope near the bluff edge changes materially. Both are a phone call now; either one is a very expensive surprise after construction starts.

INDEPENDENT SECOND OPINION (CLAUDE, ZERO EXPOSURE TO THE ABOVE)

My top independent findings — several of these are not in the published material at all:

Confirm the pipe sizing inconsistency on Oregon Avenue before storm sewer is installed (E-18 — not in published material) Sheet D-6631-15 shows a 24" HDPE storm main (215 LF at 0.30%) flowing into an 18" pipe. A larger pipe feeding a smaller pipe creates a hydraulic surcharge condition — the 24" main backs up behind the 18" restriction and floods the road surface and adjacent sites during any significant storm event. This is a design error, not a value judgment. The EOR needs to review the hydraulic calculations for Oregon Avenue and confirm whether the 18" pipe is correctly sized to handle what the 24" main delivers. If not, one of those pipes needs to change before installation. **This should be an RFI before a single foot of Oregon Avenue storm sewer goes in the ground.**

The bluff-edge outfall situation is more serious than either analysis has fully addressed (E-22) The plan set discharges stormwater at two Points of Egress on the Lake Michigan bluff — confirmed by the sheet titles "to STA 24+02 (POE)" and "to STA 28+74 (POE)" on the California Drive and Tennessee Avenue profiles. Lake Michigan bluff erosion is an active regulatory concern in Manistee County specifically — EGLE's Coastal Zone Management program has issued enforcement actions on Manistee County bluff-edge developments within the last decade. The published analysis mentions the EGLE HREA setback question. What it doesn't say: the stormwater outfall design may independently trigger a EGLE Part 325 (Great Lakes Submerged Lands) permit and a USACE Section 404 review if the discharge reaches the lake margin. That's a separate regulatory path from the Part 31 detention question, and it has its own timeline. A pre-application call with EGLE Coastal Zone — separate from the Part 31 call — should happen before detention basin design begins, because the outfall protection requirements may shape where and how the detention basin drains.

Order transformers before bid award — not a recommendation that appears in published material I mentioned this under cost, but it belongs here too as an operational recommendation. Pad-mount distribution transformers (167–250 kVA single-phase) are on 40–80 week allocation nationally. The project has approximately 5 transformer locations estimated from the site layout. If the owner waits until NTP to initiate transformer procurement, those units may not arrive until 10–20 months after groundbreaking. Every trade that depends on energized secondary — pedestal setting, panel trimming, final electrical inspection — stalls. The fix is simple: identify transformer sizes with the EOR and serving utility, get the purchase order placed before or concurrent with bid award. **This is the most actionable single schedule risk in the project and it costs nothing to mitigate if caught early.**

The service lateral plan gap is a real construction problem (E-20 — not in published material) The sanitary sewer plan set shows 8" PVC mains and manholes in detail, but I found no individual site service lateral locations or stub-up elevations anywhere in the readable plan text. For 155 sites, each needs a 4" or 6" lateral from the main to a stub-up at the pad or pedestal location, with a minimum 42" of cover for Michigan frost depth. Without lateral locations on the drawings, the utility contractor will either guess (wrong stub elevations) or stop work for an RFI — and either outcome after the main is in the ground is expensive. The EOR should add a utility rough-in plan sheet showing each site's lateral stub location, invert elevation, and cap detail before the bid goes out. Cost to add that sheet: $3,000–$8,000 in EOR time. Cost to fix mislocated laterals in a finished site: $5,000–$15,000 each.

Fire hydrant spacing needs written fire marshal confirmation before water main installation (E-10) The plan shows 8–10 hydrants across the water distribution system, but I couldn't confirm from the readable text that every site is within 400 LF of a hydrant — the full hydrant coverage map requires D-6631-04 (visual only, unreadable). If any site is more than 400 LF from a hydrant, the fire marshal won't issue a CO for that area. Retrofitting a hydrant after the water main is in and the road is paved runs $15,000–$40,000 per hydrant including pavement restoration. The fix before construction: get the EOR to produce a hydrant coverage radius map, submit it to the fire marshal, and get written approval. **This is a hold point — no water main installation until fire marshal sign-off on the hydrant layout.** The published material doesn't surface this specifically.

INDEPENDENT SECOND OPINION (CROSS-MODEL, GOOGLE GEMINI)

Before breaking ground, I highly recommend addressing the following risks and missing information:

- **Fire Flow and 6-Inch Water Mains:**
  - **OBSERVED**: Sheets 32-43 indicate the entire proposed expansion is fed by "PROPOSED 6" WATER MAIN" (C900 PVC). Fire hydrants are detailed and connected directly to these 6" lines (e.g., Sheet 40 at STA 0+37, Sheet 42 at STA 8+43).
  - **Risk**: Under most modern fire codes (like NFPA 1), a 6" water main is generally considered undersized to supply a fire hydrant, as it struggles to provide the required 1,000+ GPM fire flow without unacceptable pressure drops. While some municipalities allow 6" if heavily looped, this is a major red flag.
  - **Action**: Verify written approval from the Manistee Fire Marshal that 6" mains are sufficient for hydrants in this specific park layout. If they reject it during building permitting, you will face a costly redesign to 8" mains.
- **One-Way Road Width:**
  - **OBSERVED**: Sheet 10 specifies the "PROPOSED TYPICAL ONE-WAY ROAD CROSS SECTION" as two 7' lanes (14' total width).

- **Risk**: 14 feet is exceptionally narrow for an RV community where drivers are navigating 40-foot rigs with varying skill levels. Furthermore, the International Fire Code typically requires a 20-foot minimum clear width for fire apparatus access roads.
- **Action**: Confirm that a 14' width meets local fire apparatus access requirements and consider widening to 16'-18' simply to prevent RVs from jumping the curbs and destroying the landscaping.

- **Sandy Soils and Trench Safety:**
  - **OBSERVED**: Sheet 03 identifies the soils as "Plainfield sand" and "Covert sand".
  - **Risk**: Sand has virtually no cohesion. When excavating deep sanitary and storm trenches (some storm inverts are noted at 652 elevation against 664 rims, representing 12-foot deep trenches), the trenches will slope out significantly to meet OSHA standards (noted on Sheet 21). This will require moving much more dirt than a straight-wall trench, potentially increasing your earthwork export and surface restoration costs. Ensure your civil contractor has accounted for wide "V" trenches in their bid, not just theoretical pipe volumes.

Additional notes (not tied to a specific numbered question)

GOOGLE GEMINI CROSS-MODEL REVIEW — CONFIDENCE NOTES

- **UNKNOWN**: Existing site topography. While the text notes "EXISTING CONTOURS" and "PROPOSED CONTOURS" in the legend (Sheet 02), the raw text extraction does not provide the grading plans or cut/fill tables necessary to evaluate earthwork balancing.
- **UNKNOWN**: Individual utility lateral lengths. Sheet 31 shows a "SERVICE DETAIL" but does not quantify the linear footage of 4" or 6" PVC required to connect the 73+ pads to the mains.
- **UNKNOWN**: Primary electrical runs. Sheet 49 explicitly states "CONDUIT AND CONDUCTORS BY ELECTRIC UTILITY (VIF)" for primary feeds to transformers. Jennings Realty needs to confirm whether the local utility covers this cost or if it will be billed back to the developer as a massive line-extension fee.

PLAN ISSUE REGISTER

Status: full · Source: Plan Review - Error Register - 03A 16ft BOC Alt REV 3.md · Reviewed 2026-09-28 · 49 sheets

All 37 items transcribed in full from the real error register document (2026-09-28), parsed with the same code used for auto-ingested plans (ai_engine._parse_error_register_table).

9 Critical 7 Moderate 15 Potential 6 Low

CRITICAL (9)

C1 CRITICAL High (absent)

Index (01): no grading sheet among 49 · Sheet 1 of 49 — Cover ↗

The set has no proposed grading: no pad/FF elevations, lot spot grades, swales or proposed contours for 155 sites. Only road CL profiles (on the storm P&Ps) and existing contours (03) are shown.

CONSEQUENCE IF BUILT AS DRAWN

Lots get graded by eye. Pads can end up below the gutter or draining toward neighboring homes, and there's no overflow route at the sags.

FIELD CORRECTION

Hold mass grading. Get a sealed grading plan showing an FF/pad elevation per site, positive drainage away from each pad (suggest ≥2% for 10'), rear-lot swales ≥1%, and an overflow route at every sag. Excavator stakes pads from that plan.

WHO RESOLVES IT

EOR

PERMIT EXPOSURE

EOR letter if a grading plan exists in the permitted set. Agency notification if the MHC/Part 91 permits were issued without one and the reviewer requires it; confirm which.

DOCUMENTATION

RFI, grading sheet, pad-grade hold point, record dwgs

ROUGH COST AND SCHEDULE

Design $15–35k; 2–4 wks before mass grading

C2 CRITICAL High (geometry); Mod (alignment = CL)

17 MA101/102 STA 3+36; 19 PA101/102 STA 3+63; 20 WA101/102 STA 4+24; 18 MD101; section on 10 ·
Sheet 17 of 49 — Storm Sewer Plan And Profile Minnesota Avenue Sta 0+00 (Pob) To Sta 5+96 (Poe) ↗

The 16' BOC one-way section (10: 7'+7' lanes, 1' curb) puts the gutter at ±7' and BOC at ±8'. The CBs are labeled 10.5' LT and the leads are 21 LF, perpendicular (CB and MH stations within 1'). That puts the CB and MH at ±10.5', about 2.5' behind the back of curb. These offsets fit a 21' road, not the 16' alternate.

CONSEQUENCE IF BUILT AS DRAWN

The 7" integral HMA curb traps road water short of the inlets, so the sags pond. The structures land in the front setbacks and possibly the driveways.

FIELD CORRECTION

Hold storm work on these 4 roads. EOR relocates the inlets into the gutter at the face of curb with a frame suited to HMA curb, and moves each MH to CL or the opposite gutter (leads ≈14 LF). GC checks every stake against BOC before excavation.

WHO RESOLVES IT

EOR (RFI/field directive)

PERMIT EXPOSURE

EOR letter

DOCUMENTATION

RFI; revised 17–20; staking hold point; record

ROUGH COST AND SCHEDULE

Design $3–8k, 1–2 wks. Pre-install ≈ neutral; if installed $6–12k/structure (7 structures ≈ $40–85k)

C3 CRITICAL Moderate

Sag structures: 13 CD104/105 (rim 659.36/659.35 vs low pt 659.22 @23+68.74); 14 TA102 (660.74 vs 660.63); 15 OA102/103 (661.88/661.84 vs 661.77); 19 PA101/102 (662.46 vs 662.32); 20 WA101/102 (663.01 vs 662.86); MA/KA similar

Every inlet rim is 0.11–0.15' above the profile low-point elevation at the same station. On the crowned 2% section the gutter is about 0.14' below CL, so the grates sit about 0.25–0.30' above the flowline. Each CB rim also equals its paired MH rim, which can't happen on a crowned section.

CONSEQUENCE IF BUILT AS DRAWN

Inlets don't capture flow, so every sag ponds and ices. Castings sit proud of the pavement and take plow strikes.

FIELD CORRECTION

EOR states what the profile represents and issues a gutter flowline and rim for each inlet. Set grates 0.10–0.15' below flowline and MH lids flush. Set castings on temporary rings and do the final set after paving (21 adjustment detail).

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

Within approved plans

DOCUMENTATION

RFI; rim survey before paving; record

ROUGH COST AND SCHEDULE

$0 pre-set; $0.5–1.5k/reset after paving

C4 CRITICAL Mod-High

18 Montana Dr STA 6+03.3–6+40.3 ·

Sheet 18 of 49 — Storm Sewer Plan And Profile Montana Drive Sta 0+00 (Pob) To Sta 6+40 (Poe) ↗

The profile falls at −0.50% to a grade break at 6+29.30 (EL 661.94, no VC), then rises at +2.00% to 662.16 at Oregon. The only inlet, MD101, sits upstream at 6+03.3 with rim 662.49, 0.55' above the sag.

CONSEQUENCE IF BUILT AS DRAWN

The low point has no inlet. Water ponds about 0.2' deep and spills onto Oregon, with ice at the intersection.

FIELD CORRECTION

EOR either adds a curb inlet at 6+29 (rim ≈ gutter −0.1', lead to OA104) or re-profiles the last 40' to fall continuously into Oregon's gutter. No HMA until resolved.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

EOR letter

DOCUMENTATION

RFI; revised 18

ROUGH COST AND SCHEDULE

$4–8k; ~1 wk

C5 CRITICAL Mod-High

39 California Dr WM crossing (STA 0+00–2+31) ·

Sheet 39 of 49 — Water Main Plan And Profile California Drive Crossing Sta 0+00 To Sta 2+31 ↗

Labeled 8" san INV 658.34 and 18" storm INV 657.53 run under a WM with FL 660.1–660.5. Outside-to-outside clearance is about 12–17" to the sanitary and 9–13" to the storm. Ten States §8.8 (EGLE basis) requires 18".

CONSEQUENCE IF BUILT AS DRAWN

Separation violation. It fails the water main permit condition, and the sanitary crossing is a contamination risk.

FIELD CORRECTION

Re-profile the WM under both pipes: top of WM ≥18" below the storm's outside bottom (FL ≈655.3 or lower). Use restrained vertical bends per the 44 schedule and one full WM length centered on each crossing. Going over isn't viable (FL ≈660.9+, under 4' cover). Pothole both sewers first.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

EOR letter if 18" is achieved. Agency (EGLE) notification is unavoidable if a reduced-separation variance is used, so re-profile instead.

DOCUMENTATION

RFI; stamped profile; crossing hold point + photos

ROUGH COST AND SCHEDULE

$5–12k; 1–2 wks

C6 CRITICAL Moderate

43 Oregon WM; storm per 15 · Sheet 43 of 49 — Water Main Plan And Profile Oregon Avenue Sta 0+00 To Sta 7+05 ↗

There are 3 labeled storm crossings (INV 657.16, 656.82, 655.95), and the WM FLs range 655.8–659.4. At 657.16 and 656.82, no labeled FL gives 18" over or under. Several FLs (656.6, 656.8, 657.3) fall inside the storm pipe envelope, which is a physical clash. Sheet 43 also calls the 655.95 pipe 12", but 15 shows it as 18" (OA104 N INV 655.95).

CONSEQUENCE IF BUILT AS DRAWN

Pipes collide or the separation fails. The WM can't be laid as profiled.

FIELD CORRECTION

EOR reissues the profile with station, pipe OD and WM FL at each crossing (e.g., FL ≤ ~655.0 to pass under the 657.16 crossing). Pothole each storm crossing after storm install and before WM trenching.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

EOR letter (18" met); agency if variance

DOCUMENTATION

RFI; revised 43; hold point each crossing

ROUGH COST AND SCHEDULE

$10–25k; 1–3 wks

C7 CRITICAL Moderate

46–48 key note E3; 49 · Sheet 46 of 49 — Electrical Site Plan Southwest ↗

The E3 RV pedestals are "100A loop-fed", but no source, feeder, meter, conductor or conduit is shown. The only panels are NE and G (100A, 1ϕ, lighting). NEC 551.73 demand for 11 sites with 50A is about 53 kVA (≈220A @240V). The text shows only 7 E3 tags against 11 RV super sites (06: 16, 17, 169–177).

CONSEQUENCE IF BUILT AS DRAWN

RV sites have no power, or an undersized feeder.

FIELD CORRECTION

EOR (electrical) issues the source (≥200A service or utility meters), the 551.73 calc, feeder sizes, loop routing, and a pedestal count reconciled to 11. No RV trenching until issued.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

EOR letter + electrical permit revision

DOCUMENTATION

RFI; revised 46–49; panel schedule

ROUGH COST AND SCHEDULE

Design $3–6k; $25–60k if not in bid; utility service 8–16 wks

C8 CRITICAL (CODE) High

46–48 lighting runs "2-#10 & #12 GEC in 2" PVC"; 49 · Sheet 46 of 49 — Electrical Site Plan Southwest ↗

The EGC is #12 (and mislabeled "GEC") on #10 circuits. #10 is above the minimum for a 15/20A circuit, so NEC 250.122(B) requires a proportionally upsized EGC (#10). At 30A, #10 is the minimum anyway. No panel schedules are given to confirm breaker sizes.

CONSEQUENCE IF BUILT AS DRAWN

Fails electrical inspection.

FIELD CORRECTION

Pull a #10 Cu EGC with every #10 circuit. EOR issues NE/G panel schedules and corrects the "100A/3P" main to 2-pole for 1ϕ-3W.

WHO RESOLVES IT

Sub within spec after EOR confirms

PERMIT EXPOSURE

Within approved plans

DOCUMENTATION

RFI; record note

ROUGH COST AND SCHEDULE

Few $k; no schedule

C9 CRITICAL (MISSING INFO) Moderate

10: only section is "to apply: Minnesota, Montana, Pennsylvania, Washington" · Sheet 10 of 49 — Site Details ↗

California, Tennessee, Oregon, Kentucky, 3rd St and the emergency drive have no typical section or width. The storm geometry on those roads (12' offsets, 24 LF leads) implies 24' BOC. If the "16ft BOC" alternate applies to two-way roads, they fall below the commonly cited MHC two-way minimum (21').

CONSEQUENCE IF BUILT AS DRAWN

Paving can't be built as drawn. The wrong width means rework or a permit problem, and it also drives the storm layout (C2).

FIELD CORRECTION

EOR issues sections for every road and states which roads the alternate covers. No subgrade cut on those roads until issued.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

Within if widths match the MHC permit. **Agency notification unavoidable** if the alternate narrows any road below what was permitted; confirm what was permitted first.

DOCUMENTATION

RFI; revised 10

ROUGH COST AND SCHEDULE

Design $2–5k; ~1 wk

MODERATE (7)

M1 MODERATE High

13 CD104→CD105; 16 KA105→KA104 ·
Sheet 13 of 49 — Storm Sewer Plan And Profile California Drive Sta 20+00 To Sta 24+02 (Poe) ↗

Two CB leads have negative grade: CD104 out 655.31 → CD105 in 655.37 (–0.06'/24'), and KA105 out 656.23 → KA104 N in 656.35 (–0.12'/24').

CONSEQUENCE IF BUILT AS DRAWN

Standing water and sediment in the leads; fails line-and-grade inspection.

FIELD CORRECTION

Raise the CB outlets to CD104 ≥655.49 and KA105 ≥656.47 (0.5%), or lower the receiving inverts, as the EOR confirms. Shoot inverts before backfill.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

Within

DOCUMENTATION

RFI; invert survey

ROUGH COST AND SCHEDULE

~$0 pre-install; $1–3k each to relay

M2 MODERATE High

14 TA102 (24" in/30" out all 653.01), TA-103 (653.46 in/out); 16 KA102 (all 655.20), KA101; 19 PA102 (12"/15" both 657.10); 12 CD102; 15 OA102/105; 20 WA102 ·

Sheet 14 of 49 — Storm Sewer Plan And Profile Tennessee Avenue Sta 19+00 To Sta 28+74 (Poe) ↗

Several structures have zero drop, and size increases are matched at the invert, not the crown. At TA102 the 30" outlet crown is 0.5' above the 24" inlet crown.

CONSEQUENCE IF BUILT AS DRAWN

Backwater, lost capacity, sediment.

FIELD CORRECTION

Use a 0.10' minimum drop through structures and match crowns (or 0.8D) at size changes (e.g., TA102 30" out at ≈652.51). EOR checks the downstream cascade and cover.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

Within

DOCUMENTATION

RFI; invert hold point

ROUGH COST AND SCHEDULE

$0–5k design

M3 MODERATE Moderate

12 CD102 387 LF 12" @0.26%; 15 OA106→105 110 LF @0.26%; 13 CD105 outlet 12" @0.26% ·

Sheet 12 of 49 — Storm Sewer Plan And Profile California Drive Sta 12+00 To Sta 20+00 ↗

12" HDPE at 0.26% runs about 2.5 fps full (≈2.0 cfs), below the ~3 fps self-cleansing target. CD105 serves the lowest road sag (659.22).

CONSEQUENCE IF BUILT AS DRAWN

Sand sedimentation, recurring jetting, and low capacity at the critical sag.

FIELD CORRECTION

Upsize to 15" at the same inverts (≈2.9 fps, ≈3.6 cfs) or steepen where cover allows. Laser-set these runs with zero negative tolerance.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

Within

DOCUMENTATION

RFI; record

ROUGH COST AND SCHEDULE

≈$5–10k

M4 MODERATE Moderate

All structure callouts "MDOT Cover C" (12–20) vs 21 CB detail "EJIW 7045 / M1 grate (curb & gutter)"

Two different castings are specified. The 7045 is a concrete C&G frame, but the roads have integral HMA curb, and the grate vs solid choice for the 48" structures is unclear.

CONSEQUENCE IF BUILT AS DRAWN

Wrong castings get ordered, fit poorly, and capture poorly.

FIELD CORRECTION

EOR issues a casting schedule by structure (gutter grate vs solid lid, HMA-curb-compatible frame). Castings go through submittal before ordering.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

Within

DOCUMENTATION

RFI; submittal

ROUGH COST AND SCHEDULE

$0–300/casting

M5 MODERATE High (absent)

31 (municipal service detail only); 44; 32–43 · Sheet 31 of 49 — Sanitary Sewer Details ↗

There are no MH/RV site connection details: water service size/material, curb stop, frost-protected riser, sewer riser/cleanout location vs pad. The 31 detail is a municipal ROW stub (10' min depth, plug + witness marker), and the WM sheets call out no service taps.

CONSEQUENCE IF BUILT AS DRAWN

Services get built inconsistently, with freeze-ups and misplaced risers. This is a large bid-scope gap across 155 sites.

FIELD CORRECTION

EOR issues a site-utility detail per site type (riser locations, depths, frost protection, curb stop boxes). The utility sub prices per site from it.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

Within

DOCUMENTATION

RFI; new detail; per-site record

ROUGH COST AND SCHEDULE

$1–2.5k/site ≈ $0.15–0.4M if not carried

M6 MODERATE Moderate

46–48 notes E1/E2 · Sheet 46 of 49 — Electrical Site Plan Southwest ↗

MH services say only "100A/200A 1ϕ … meter pedestal to utility standards". There's no main breaker/disconnect, no pedestal type, and no home feeder. NEC 550.32 requires the disconnect at service equipment near the home.

CONSEQUENCE IF BUILT AS DRAWN

Meter-only pedestals get bought, and inspection fails.

FIELD CORRECTION

Specify utility-approved meter/main pedestals (100A/200A main) and who furnishes the home feeder. Submittal required.

WHO RESOLVES IT

EOR RFI

PERMIT EXPOSURE

Within

DOCUMENTATION

RFI; submittal

ROUGH COST AND SCHEDULE

$300–700/site ≈ $45–100k if not carried

M7 MODERATE High

06/08/11 "PR 32'×40' maintenance garage/building", 11 "PR pavilion" vs 47/49 note 15 "future" garage · Sheet 6 of 49 — Parcel Types ↗

Civil sheets show the garage and pavilion as proposed; electrical shows the garage as future. No water/sewer/electric stubs, building plans or pavilion detail are in the set.

CONSEQUENCE IF BUILT AS DRAWN

Scope gap; stubs later get cut through new paving.

FIELD CORRECTION

Owner decides in or out of scope. If in, EOR issues stubs (water, sewer, power from panel G) before paving.

WHO RESOLVES IT

Owner + EOR

PERMIT EXPOSURE

Within

DOCUMENTATION

Scope clarification / CO

ROUGH COST AND SCHEDULE

Stubs $10–25k

POTENTIAL (15)

P1 POTENTIAL —

03 SESC; soils table · Sheet 3 of 49 — Sesc Plan ↗

Plainfield sand 67F (35–70% slopes) is mapped on site, next to Manistee Lake. The SESC key has only seed/mulch, silt fence, inlet protection and one stone access, with no slope blanket or diversion. Resolves: disturbance limits vs 67F areas; Part 91 permit conditions.

CONSEQUENCE IF BUILT AS DRAWN

Slope erosion and sediment to the lake.

FIELD CORRECTION

If 67F areas are disturbed, add blankets and diversions and add a stone access per phase.

WHO RESOLVES IT

GC + EOR

PERMIT EXPOSURE

Within

DOCUMENTATION

SESC log

ROUGH COST AND SCHEDULE

$2–10k

P2 POTENTIAL —

07–09 legend "Relocate by others (REL-BO)" gas/electric/fiber/tele/water · Sheet 7 of 49 — Parcel Dimensions Southwest ↗

Existing utilities are flagged for relocation by others; owners and dates aren't identifiable. Resolves: owner relocation schedule.

CONSEQUENCE IF BUILT AS DRAWN

Crews idle or strike live lines.

FIELD CORRECTION

Get relocation dates before clearing; MISS DIG plus potholing.

WHO RESOLVES IT
GC/Owner

PERMIT EXPOSURE
Within

DOCUMENTATION
Log

ROUGH COST AND SCHEDULE
Schedule risk

P3 POTENTIAL —

25 EX MH #2 (rim 662.86, IE S 647.76 / E 647.56) ·
Sheet 25 of 49 — Sanitary Sewer Plan And Profile Oregon Avenue Sta 9+00 To Sta 18+63 (Poe) ↗

155 new sites discharge to an existing 8" line. Downstream capacity isn't shown, and OA3 out 648.24 leaves only 0.48' of fall to the core at 647.76. Resolves: Part 41 permit basis; field survey of EX MH #2.

CONSEQUENCE IF BUILT AS DRAWN
Surcharge, or a connection that doesn't fall.

FIELD CORRECTION
Survey EX MH #2 before setting any Oregon sanitary.

WHO RESOLVES IT
EOR

PERMIT EXPOSURE
Agency unavoidable if the permitted flow/site count differs

DOCUMENTATION
Survey

ROUGH COST AND SCHEDULE
$1–2k

P4 POTENTIAL —

14/24 TA102 23+14.5 vs TA4 23+23.0; TA-103 25+37.6 vs TA5 25+47.4; 12/22 CD102 18+12.4 vs CD3 18+01.7 ·
Sheet 14 of 49 — Storm Sewer Plan And Profile Tennessee Avenue Sta 19+00 To Sta 28+74 (Poe) ↗

Storm and sanitary MHs sit 8.5–10.7' apart on the same stationing with no MH offsets given. At TA4 the sanitary invert (655.60–655.70) is at the 30"/24" storm crown elevation. Resolves: horizontal offsets on the graphics.

CONSEQUENCE IF BUILT AS DRAWN
Hard vertical conflict if the lines are collinear or cross nearby.

FIELD CORRECTION
Stake both systems together before installing the sanitary; pothole.

WHO RESOLVES IT
EOR

PERMIT EXPOSURE
—

DOCUMENTATION
Staking hold

ROUGH COST AND SCHEDULE
—

P5 POTENTIAL —

11–21; EX CB 5018 (36"), 36" at 653.83 · Sheet 11 of 49 — Overall Utility Plan ↗

There are no drainage calcs, area map, detention or overflow routes. The new area ties to the existing 36" system, and the NE sag is served by 12" @0.26%. Resolves: EOR calcs and the approval they went to (township/drain commissioner).

CONSEQUENCE IF BUILT AS DRAWN

Under-capacity or unpermitted discharge.

FIELD CORRECTION

Get the calcs; define an emergency overflow at the NE sag.

WHO RESOLVES IT

EOR

PERMIT EXPOSURE

Agency unavoidable if volume/discharge differs from approval

DOCUMENTATION

—

ROUGH COST AND SCHEDULE

—

P6 POTENTIAL —

32 Kentucky (WM FL 652.1–654.8 vs san 652.00–653.28); 43 Oregon ·
Sheet 32 of 49 — Water Main Plan And Profile Kentucky Avenue Sta 0+00 To 10+60 ↗

The WM runs parallel to the sanitary. On Kentucky it drops to or below the sanitary crown, so 10' outside-to-outside horizontal separation (Ten States 8.8.2) is the only compliant option. Resolves: offsets.

CONSEQUENCE IF BUILT AS DRAWN

Separation violation.

FIELD CORRECTION

Measure 10' at staking. If it's under 10', EOR redesigns; this can't be waived in the field.

WHO RESOLVES IT

EOR

PERMIT EXPOSURE

Agency unavoidable if <10'

DOCUMENTATION

Photo log

ROUGH COST AND SCHEDULE

—

P7 POTENTIAL —

36 (san 657.86 "18" MIN", FL 659.9–660.3); 40 (storm 655.66); 42 (storm 655.99, san 652.00); cover at 39/43 ·
Sheet 36 of 49 — Water Main Plan And Profile Rear Lot Washington Avenue Sta 0+00 To Sta 8+17 ↗

FLs can't be tied to crossing stations from the text. Sheet 36 meets 18" only if FL ≥660.06 at the crossing (659.9 gives 16"). Some FLs may leave under 5' of cover. Resolves: graphics plus the water utility's cover standard.

CONSEQUENCE IF BUILT AS DRAWN

Separation or frost violation.

FIELD CORRECTION

Expose and measure every crossing; verify cover.

WHO RESOLVES IT

EOR

PERMIT EXPOSURE

EOR letter

DOCUMENTATION

Photo log

ROUGH COST AND SCHEDULE

—

P8 POTENTIAL —

32 Kentucky 10+60 6"×6" tapping sleeve (EX FL 660.2 "LOCATE"); 38/39 remove existing plugs ·
Sheet 32 of 49 — Water Main Plan And Profile Kentucky Avenue Sta 0+00 To 10+60 ↗

The existing main's material and depth are unknown, and a size-on-size tap needs a full-body sleeve rated for that pipe.

CONSEQUENCE IF BUILT AS DRAWN

Wrong sleeve or failed tap.

FIELD CORRECTION

Pothole both before ordering; utility operator present; shutdown plan for plug removal.

WHO RESOLVES IT

GC/Owner

PERMIT EXPOSURE

Within

DOCUMENTATION

Hold point

ROUGH COST AND SCHEDULE

$1–3k; sleeve 2–6 wks

P9 POTENTIAL —

32–44 · Sheet 32 of 49 — Water Main Plan And Profile Kentucky Avenue Sta 0+00 To 10+60 ↗

All mains are 6" C900 with no fire-flow requirement, test or calc. No hydrants on Kentucky (1,060 LF) or the Minnesota/Montana/Pennsylvania rear-lot mains; some hydrants sit on rear-lot mains behind homes; no hydrant assembly detail on 44. Resolves: fire authority's flow and spacing requirements.

CONSEQUENCE IF BUILT AS DRAWN

Inadequate fire protection.

FIELD CORRECTION

Get a fire authority letter before WM install; add hydrants or upsize per EOR.

WHO RESOLVES IT

EOR + fire

PERMIT EXPOSURE

Agency unavoidable if the fire layout changes

DOCUMENTATION

—

ROUGH COST AND SCHEDULE

Hydrant ≈$6–9k ea

P10 POTENTIAL —

45–49 · Sheet 45 of 49 — Electrical Site Plan Overall ↗

Electric distribution is "by utility (VIF)" and not shown, so transformer kVA, voltage drop and joint-trench conflicts can't be checked. 100A single-wide services may be small for all-electric homes. E1/E2 tags in the text (~100 on overlapping sheets) need reconciling to 144 MH sites. No gas service is shown.

CONSEQUENCE IF BUILT AS DRAWN

Late utility conflicts; under-served sites.

FIELD CORRECTION

Overlay the utility design on the civil before trenching; reconcile tags to 06.

WHO RESOLVES IT

Utility + EOR

PERMIT EXPOSURE

Within

DOCUMENTATION

—

ROUGH COST AND SCHEDULE

Utility lead time

P11 POTENTIAL —

10, 04, 08 · Sheet 10 of 49 — Site Details ↗

The 16' BOC one-way roads (14' drivable) meet the commonly cited MHC one-way minimum (13') only with no parking, and 04 shows no No-Parking signs. The fire authority may require 20' (IFC 503.2.1). There are no curb radii, no MH-transport or RV back-in templates (sites 169–177 off Washington), and no emergency drive section.

CONSEQUENCE IF BUILT AS DRAWN

Blocked apparatus and delivery; damaged curb.

FIELD CORRECTION

Add No-Parking signs, run turning templates, and widen radii before curb goes in.

WHO RESOLVES IT

EOR + fire

PERMIT EXPOSURE

Agency if widths change

DOCUMENTATION

—

ROUGH COST AND SCHEDULE

Signs $150–300 ea

P12 POTENTIAL —

06, 10 · Sheet 6 of 49 — Parcel Types ↗

The licensing path for the 11 RV super sites (MHC vs. EGLE Part 125 campground) isn't stated, and per-site RV water/sewer hookups aren't shown.

CONSEQUENCE IF BUILT AS DRAWN

Unpermitted sites.

FIELD CORRECTION

Confirm permits and hookup design before RV pads.

WHO RESOLVES IT

Owner/EOR

PERMIT EXPOSURE

Agency unavoidable if not covered

DOCUMENTATION

—

ROUGH COST AND SCHEDULE

—

P13 POTENTIAL —

33–42 rear-lot WMs; 10 lot details (garages, 8×10 shed slabs) ·
Sheet 33 of 49 — Water Main Plan And Profile Rear Lot Minnesota Avenue Sta 0+00 To Sta 4+45 ↗

The lot details don't show a WM easement or no-build zone.

CONSEQUENCE IF BUILT AS DRAWN

Sheds or garages set over the main.

FIELD CORRECTION

Add an easement line to the lot details and set-up rules.

WHO RESOLVES IT

EOR/Owner

PERMIT EXPOSURE

Within

DOCUMENTATION

Record

ROUGH COST AND SCHEDULE

—

P14 POTENTIAL —

21, 22–31, 32–44 · Sheet 21 of 49 — Storm Sewer Details ↗

Not on the plans: sanitary PVC class and trench/bedding detail, C900 DR, tracer wire, the "Type A/B granular" definitions, and the basis of the restraint schedule. Resolves: project manual.

CONSEQUENCE IF BUILT AS DRAWN

Wrong materials.

FIELD CORRECTION

If absent, EOR adds SDR 26 at >12' cover (OA3 ≈15'), C900 DR18, and tracer wire.

WHO RESOLVES IT

EOR

PERMIT EXPOSURE

Within

DOCUMENTATION

Submittals

ROUGH COST AND SCHEDULE

Low

P15 POTENTIAL —

04 (930 LF privacy fence, Class A buffer alt 3); 02 barrier-free legend; 11 pavilion · Sheet 4 of 49 — Site Overall ↗

There's no fence detail even though the buffer is a zoning condition, and no barrier-free parking or route to the pavilion.

CONSEQUENCE IF BUILT AS DRAWN

Zoning or ADA noncompliance.

FIELD CORRECTION

Get the zoning buffer spec; add an accessible route if the pavilion is built.

WHO RESOLVES IT

Owner/EOR

PERMIT EXPOSURE

Agency if the zoning spec isn't met

DOCUMENTATION

—

ROUGH COST AND SCHEDULE

—

LOW (6)

L1 LOW —

All title blocks

The file is named "REV 3 — SEALED", but every drawing log shows only "8/20/26 Issued for Bidding". Dates read JULY 2026 (01–10, 44–49) and AUGUST 2026 (12–43).

CONSEQUENCE IF BUILT AS DRAWN

Building from a bid set.

FIELD CORRECTION

Get written confirmation of the governing IFC set.

WHO RESOLVES IT

Owner/EOR

PERMIT EXPOSURE

—

DOCUMENTATION

Transmittal

ROUGH COST AND SCHEDULE

—

L2 LOW —

01 vs 14, 16, 24, 26, 34, 39 · Sheet 1 of 49 — Cover ↗

Index vs title mismatches: 18+00 vs 19+00; "0+00 (POE)"; 5+40 vs 5+01; 39 title "STA 9+00 to 2+31" and code SW8.

CONSEQUENCE IF BUILT AS DRAWN

Confusion.

FIELD CORRECTION

Tell the crew; fix on record set.

WHO RESOLVES IT

EOR

PERMIT EXPOSURE

—

DOCUMENTATION

Record

ROUGH COST AND SCHEDULE

—

L3 LOW —

32, 33, 36–40, 42, 43 · Sheet 32 of 49 — Water Main Plan And Profile Kentucky Avenue Sta 0+00 To 10+60 ↗

WM matchline references point to sanitary sheet numbers (22–33), apparently from an old numbering.

CONSEQUENCE IF BUILT AS DRAWN

Crews go to the wrong sheet.

FIELD CORRECTION

Same as L2.

WHO RESOLVES IT

EOR

PERMIT EXPOSURE

—

DOCUMENTATION

Record

ROUGH COST AND SCHEDULE

—

L4 LOW —

12–20, 39, 25/30 · Sheet 12 of 49 — Storm Sewer Plan And Profile California Drive Sta 12+00 To Sta 20+00 ↗

Mislabels: CB leads read "12" PVC SAN SWR"; the 24" KA101→TA-103 storm reads "24" PVC SAN SWR @0.20%" (14); 39's "18" STM 657.53" matches no 18" on 12–13; CB 5018 is at "STA 9+47.1", past Washington POE 6+81; OA2 has no S inlet invert from OA1.

CONSEQUENCE IF BUILT AS DRAWN

Procurement errors.

FIELD CORRECTION

Order storm leads as storm HDPE/PVC per spec; confirm OA2 S invert (≥648.85).

WHO RESOLVES IT

EOR

PERMIT EXPOSURE

—

DOCUMENTATION

Record

ROUGH COST AND SCHEDULE

—

L5 LOW —

03–08, 32, 40–42 · Sheet 3 of 49 — Sesc Plan ↗

Colorado Drive vs Avenue; OREGAON; TENESSEE; PENNSLYVANIA.

CONSEQUENCE IF BUILT AS DRAWN

Signage errors.

FIELD CORRECTION

Use correct names on signs.

WHO RESOLVES IT

GC

PERMIT EXPOSURE

—

DOCUMENTATION

—

ROUGH COST AND SCHEDULE

—

L6 LOW —

03, 10, 21, 31, 36, 40, 44, 49 · Sheet 3 of 49 — Sesc Plan ↗

Typos: "5EL/4EL" (MDOT 5E1/4E1); "8' precast slab" (8"); witness marker 12' vs 12"; "#1/0 copperweld ground rod"; "100A/3P" on 1ϕ; undefined asterisk "PG-05,07*"; "6": HYDRANT"/"VSALVE"; SESC note 7 missing; HDPE DR-11 details on a C900 job; undefined abbreviations (VIF, MCB, RM, GEC, POB/POE, PVI, DEFL).

CONSEQUENCE IF BUILT AS DRAWN

None if the crew is told.

FIELD CORRECTION

Brief at pre-con.

WHO RESOLVES IT

EOR

PERMIT EXPOSURE

—

DOCUMENTATION

Record

ROUGH COST AND SCHEDULE

—

DO THESE NEXT

- ✓ Reconcile the two new long-term-hold estimates ($11.87M vs $9.60M) with Jay/Marc before either becomes the real working budget
- ✓ Compare Grex Dev's real contractor bids (due Oct 6, comparison expected Oct 7) against these estimates
- ✓ Confirm Consumers Energy will size transformers for 200A service on every manufactured-home lot
- ✓ Get a sealed grading plan -- none exists for any of the 155 sites (Critical finding C1)
- ✓ Confirm with Spicer which road width the actual MHC construction permit was issued on
- ✓ Decide on Jay's one-sub utility-trenching idea once Consumers Energy and DTE respond
- ✓ Confirm Bluffs' general liability actually meets the lender's $2M/$5M minimum (Byline Schedule 5.5) -- the company's own insurance review (Sep 2026) found current coverage may be only $1M per occurrence with a $50,000 deductible, and the umbrella policy doesn't name Bluffs at all
- ✓ Get Bluffs' workers comp bound, not just quoted -- the insurance review found no current policy on file for a site under active construction

SOURCE DOCUMENTS

Full text of each document is below, in the same order as this list -- nothing minimized or behind a toggle. The live source files (PDF/Excel/Word) are linked in the Files section further down.

- **Class 4 Estimate (full report)** [class4]
- **Cost & Value Engineering Memo (full)** [cost_ve]
- **Ideas and ROI (full, sourced)** [revenue] — `revenue_within_approved_plans.md`
- **Additional Ideas & Risks (deep research)** [research_ideas]

- **Long-Term Hold -- Class 4 Estimate Rev 1 (full, 10/2/26)** [long_term_hold_class4]
- **Long-Term Hold -- Cost & VE Memo Rev 2 (full, 10/2/26)** [long_term_hold_cost_ve]
- **Long-Term Hold -- Revenue & VE Within Approved Plans (full, 10/2/26)** [long_term_hold_revenue]
- **Long-Term Hold Addendum -- Register Changes & Next Steps (10/2/26)** [long_term_hold_addendum]

SOURCE DOCUMENTS (FULL TEXT)

Class 4 Estimate (full report) [class4]

Preliminary Cost Estimate — AACE Class 4 (Non-Binding)

The Bluffs on Manistee Lake — North Expansion | Manistee Township, Manistee County, Michigan

Owner: M Sites, LLC | Engineer: Spicer Group (Proj. 139471SG2025) | Plan set D-6631, 49 sheets, 16-ft BOC Alternate, Rev 3, Issued for Bidding 8/20/26 | Estimate date 9/28/2026 | Pricing basis 2026 Q3

Companion workbook: Bluffs_North_Expansion_Class4_Estimate.xlsx (live formulas; the workbook governs if figures differ). This is a preliminary budget for planning only, not a bid or guaranteed price.

Executive Summary

Metric	Value
New sites	155 (144 manufactured-home lots + 11 RV super sites)
Total direct hard cost	$6,503,858 ($41,960 / site)
Hard cost incl. 12% contingency	$7,284,321 ($46,996 / site)
Soft costs incl. financing + 10% contingency	$1,687,112
TOTAL PROJECT COST (point estimate)	**$8,971,433 ($57,880 / site; ~$271,900 / gross acre)**
Class 4 range (–20% / +30%)	$7.18M – $11.66M ($46,300 – $75,200 / site)
Phase 1 (placeholder 60 sites)	$5.39M (~$89,800 / Phase 1 site — shared infrastructure front-loaded)

Bottom line: at about $57,900 per site all-in, the project sits at the low end of the ground-up MHP range and in the middle of the range for expanding an existing community. That makes sense for an expansion that uses the existing Bluffs amenities (clubhouse, pool) and extends existing 6-inch water, gravity sewer and storm systems. It needs no lift station, treatment plant, or detention pond, and the soils are easy-digging sand.

Where the money is: Division 33 utilities are 36% of hard cost, Division 26 electrical 19%, and Division 01 general requirements 13%. The biggest open risks are not construction quantities. They are (1) whether the system receiving the sewer and water has enough capacity and what fees it charges, and (2) what Consumers Energy will charge for the line extension (CIAC). Neither appears on the drawings, and each could move the total by $0.5–1.2M.

1. Scope of Work Audit

Location and project identity

The cover sheet places the project east of Manistee Lake near Eastlake Rd, Caberfae Hwy and M-55 in Manistee County, Michigan. The existing community is at 956 Tennessee Ave, Manistee Township. It is a 55+ MH/RV community that already has a clubhouse, pool, hot tub and pickleball courts. The location can be identified, so a regional factor was applied instead of a national default.

Site count and density

Type	Count	Basis	Confidence
MH double-wide parcels	63	Sheet 06 tables; avg ~7,900 SF lot, 24×26 garage detail	Moderate (split)

Type	Count	Basis	Confidence
MH single-wide parcels	81	Sheet 06 tables; avg ~6,800 SF lot, 20×20 garage detail	Moderate (split)
RV super sites (back-in, full hookup)	11	Lots 16, 17, 169–177; concrete RV pad + patio; 100A 50/30/20A pedestal	High
Total new sites	155	Lots 16–55, 77–103, 169–256	High
Existing-lot upgrades (not new)	15	RV pads 132–135; approaches 136–138; pad + approach 148–155	High
ADA sites / pull-through RV	None identified	Barrier-free symbol in legend only	Moderate

Net lot area is 1,137,598 SF (26.1 ac), the sum of all 155 parcel areas on Sheet 06. The plans do not give a gross site area, so I assumed ~33 ac (range 31–36). That gives about 4.7 sites/gross acre and 5.9 sites/net lot acre. The typical-lot details show 12-ft front setbacks, 10- and 20-ft separations to adjacent structures, and 25-ft vision-clearance triangles.

Roadway network

New private streets total about 6,463 LF centerline (1.22 mi): California 1,202, Tennessee 974, Oregon 963, Kentucky 720, Minnesota 596, Montana 640, Pennsylvania 687 and Washington 681. Minnesota, Montana, Pennsylvania and Washington are one-way (R6-1R/R5-1 signage). The 16-ft back-of-curb alternate section is two 7-ft lanes with 7" integral HMA curb. The pavement is 1.5" 5EL wearing course over 1.5" 4EL leveling course on 8" modified 21AA aggregate. Curb catch basins (EJIW 7045/M1) confirm the section is curbed. The set also shows paved parking, a pedestrian sidewalk and an emergency access drive to the maintenance garage, but none of these is dimensioned.

Water

About 7,609 LF of 6" C900 PVC runs mostly in rear-lot easements. It ties to the existing 6" community main at three points (tapping sleeve at Kentucky 10+60 and two plug removals on California). The drawings show 10 hydrants, so fire protection by hydrant is intended. The plans do not say whether the water comes from municipal supply or community wells. The 6" mains carrying hydrants and the dead ends should be checked against a fire-flow test.

Wastewater — treatment approach (flagged)

This is conventional gravity collection that feeds an existing system. About 4,900 LF of 8" PVC at 0.40%, 9–16 ft deep, with about 30 manholes, drains to existing MH #2 on Oregon Ave (core and Kor-N-Seal boot). The drawings show no lift station, force main, package WWTP or septic drainfield. That removes the usual single largest cost swing on these projects, but only if the downstream system has capacity. Neither the plans nor public sources I could reach say whether the existing Bluffs collection system discharges to a municipal plant (City of Manistee) or to a private/community system, or what capacity is left. Confirm this before relying on the $0 treatment carry.

Stormwater

About 2,770 LF of HDPE storm sewer (12" to 36") with about 30 structures ties into the existing 36" storm system at CB #5018 and at the TA102/OA102 outlets. No detention or retention basin, culverts or swales are shown. The SESC plan (Sheet 03) follows the Michigan Part 91 / MACDC APA manual: silt fence, stone construction access, inlet protection, and seeding within 72 hours. Earth disturbance over 1 acre also triggers EGLE NPDES Notice of Coverage. SSURGO soils are Plainfield and Covert sands, hydrologic group A, not hydric, and very good for infiltration.

Electrical

The utility (Consumers Energy territory) provides primary, transformers and secondary conduit and conductors, noted as "by electric utility (VIF)". There is no one-line diagram, transformer schedule or kVA shown. The

developer installs individual utility meter pedestals and service-entrance conduit: key note E1 is 100A 1Ø and E2 is 200A 1Ø, which means individual utility metering for MH lots. The 11 RV super sites get 100A loop-fed pedestals (NEMA 3R, one each 14-50R, TT-30R and 5-20R GFCI). Site lighting is about 20 SL1 poles (25-ft, 80W LED) fed from two 100A 120/240V panels on exterior racks (NE and G) with wireless controls. Each lot also gets a solar lamp post. The extracted E1/E2 tags account for only about 100 of the 144 MH lots, so I split the pedestals pro rata (105 at 100A, 39 at 200A).

Gas, vertical structures and amenities

- Gas: the legend shows gas lines, but there is no gas distribution sheet. Excluded; assumed utility-installed.
- Maintenance garage 32×40 (1,280 SF). Sheets 06/08 show it as proposed and Sheet 47 labels it "FUTURE". No building plans; assumed post-frame or PEMB.
- Pavilion: noted on Sheet 11 with no size. Assumed 24×40 (960 SF) open timber.
- No new clubhouse, bathhouse/laundry, office, pool or dump station. The existing community amenities serve the expansion.
- 930 LF privacy fence; 20-ft Class A buffer (Alternate 3); existing trees to remain; MDOT traffic and street-name signs. Mail kiosks, playground, dog park, storm shelter, dumpster enclosures and entry gates are not shown.
- Typical lots show garages (20×20 SW / 24×26 DW) and 8×10 shed slabs. Garages and homes are excluded as home-sale scope. Shed slabs are included and should be verified.

Engineering and regulatory notes

- Soils are excessively drained sands with 0–12% slopes on the developable area and 35–70% (67F) at the bluff edge. No geotechnical report is referenced; only the SSURGO table.
- No floodplain or wetland designation is noted on the plans. The site is roughly 70–80 ft above Manistee Lake, so floodplain is unlikely, but the FIRM panel should be confirmed. The wetland symbol appears in the legend only.
- Work in the ROW follows Manistee County Road Commission standards. Site lighting must comply with Michigan R 125.1908 and the MI Energy Code (IECC 2021).
- Status: sealed and issued for bidding, Rev 3. The plans do not list conditions of approval. The 49-sheet index has no grading plan, which is a notable gap.

2. Quantity Take-Off Summary

The full QTO, with the sheet reference and confidence for each line, is on the workbook QTO tab. Highlights:

Category	Quantity	Confidence
Clearing & grubbing	24 ac (range 15–30)	Low
Topsoil strip 6" / respread 3"	26,620 CY / 9,280 CY	Moderate
Mass grading (balanced)	32,000 CY (range 20k–50k); no grading plan	Low
HMA 3" / 8" 21AA base	10,054 SY / 12,926 SY	High
Integral HMA curb	12,926 LF	Moderate
Parking / emergency drive (allowance)	900 SY / 360 SY	Low
Concrete: RV pads, patios, approaches, shed slabs, walk	19,320 / 2,640 / 3,300 / 11,520 / 12,500 SF (~49,300 SF, ~660 CY)	Low–Moderate
6" C900 water main	7,609 LF; 38 valves; 10 hydrants; 45 fittings; 155 services	High (main)
8" PVC gravity sanitary	4,900 LF (4,234 legible); 30 MH; 5,425 LF 6" leads; 40 risers	Moderate
HDPE storm	12" 1,100 / 15" 333 / 18" 533 / 24" 650 / 30–36" 150 LF; 20 × 48" + 10 × 24" structures	Moderate
Electrical	105 × 100A + 39 × 200A meter pedestals; 11 RV pedestals; 20 light poles; 3,600 LF lighting circuit; 7,750 LF service conduit; 2 panels	Low–Moderate
Transformers	Utility-owned; est. 16–26 pad-mounts at 25–50 kVA (not shown)	Low

Category	Quantity	Confidence
Buildings	Garage 1,280 SF; pavilion 960 SF	Low
Lawn seeding / silt fence	23 ac / 4,500 LF; 30 inlet protections; 3 construction entrances	Moderate

3. Cost Breakdown Structure

Regional factor: The baseline is 2026 Q3 national-average installed unit costs (U.S. 30-city average = 1.00). These are estimator judgment calibrated to recent Midwest bid tabulations, not a published price book. Manistee is not an indexed city. Nearby northern-lower-Michigan locations (Traverse City, Cadillac) have historically run about 0.88–0.92 composite. I applied 0.95 to sitework, utilities and Div 01/02, 0.94 to electrical and plumbing, 0.93 to concrete and 0.92 to buildings. The factors are set above the local wage index because of remote mobilization, a short May–October paving season and a thin bidder pool. Each factor is an editable input.

CSI Division	Unit-cost basis (national, before factor)	Cost	$/site	Conf.
01 General Requirements	Mob 4%; GCs $42k/mo × 9; layout $60k; SWPPP $3k/mo; testing $40k; bonds/ins 1.75%	$831,694	$5,366	Moderate
02 Existing Conditions	Clear & grub $7,000/ac; removals/tie-ins LS	$197,600	$1,275	Low
03 Concrete	RV pads $9.50/SF; patios $8.00; approaches $12.00; shed slabs $9.00; walk $8.50	$422,397	$2,725	Low
06–09 Amenity Buildings	Garage $110/SF; pavilion $75/SF	$195,776	$1,263	Low
22 Plumbing	RV riser & frost-proof hydrant $1,500/site	$15,510	$100	Moderate
26 Electrical	Meter ped. $1,750 (100A) / $2,450 (200A); RV ped. $2,900; light pole $7,200; circuits $24/LF; CIAC allowance $3,000/site	$1,208,925	$7,800	Low
31 Earthwork	Strip $3.50/CY; mass grade $5.00/CY; subgrade $1.75/SY; pad grade $950/lot; SESC by LF/EA	$477,324	$3,080	Low
32 Exterior Improvements	HMA $27/SY; 21AA $12.50/SY; curb $10/LF; fence $52/LF; seed $3,000/ac; buffer $60k	$797,843	$5,147	Moderate
33 Utilities	6" C900 $64/LF; valve $2,600; hydrant $9,000; 8" san. $105/LF; MH $8,000; storm $50–150/LF; CB $2,800–5,200	$2,356,789	$15,205	Mod–High
TOTAL DIRECT HARD COST		**$6,503,858**	**$41,960**	

Electrical transformers are priced as a per-site utility CIAC rather than $/kVA because the plans assign transformers to the utility. If the developer ends up buying pad-mounts, budget about $90–$140/kVA installed for 25–50 kVA single-phase units.

Soft costs

Item	Basis	Full buildout
Civil engineering & CA	5.0% of hard (much likely already incurred)	$325,193
Surveying, staking, as-builts	1.25% of hard	$81,298
Geotechnical	Line item (none referenced)	$28,000
Permits & plan review	EGLE Part 41/Act 399, Part 91, NPDES, LARA MHC, township	$45,000
Sewer/water tap & capacity fees	$3,000/site allowance (provider unknown)	$465,000
Legal / entitlement	Line item	$65,000
Construction management / owner's rep	3.0% of hard	$195,116
Builder's risk / owner insurance	0.35% of hard	$22,764
Financing: interest carry	65% LTC × 8.25% × 55% avg balance × 12 mo	$251,042
Financing: origination	1.0% of loan	$55,326
TOTAL SOFT COSTS	**~23.6% of direct hard**	**$1,533,738**

Roll-up and benchmark

Measure	This project	Typical range	Position
Total $/site	$57,880	Ground-up MHP $55k–$95k	Low end
Total $/site	$57,880	MHP expansion $40k–$70k	Mid-range
Total $/site	$57,880	RV full-hookup $30k–$70k	Mid-range
Total $/gross acre	~$271,900	$250k–$500k	Low end

The benchmark ranges are estimator judgment for 2025–26 Midwest horizontal development excluding homes, not a published survey. The project is in line, not an outlier. It comes in lower than a ground-up community mainly because there is no new clubhouse, pool, bathhouse, treatment plant or detention pond.

4. Contingency & Risk Allowances

Recommendation: 12% hard-cost contingency ($780,463) + 10% soft-cost contingency ($153,374). The IFB plan-and-profile sheets define the horizontal work well, and Div 32/33 quantities are close to Class 3 quality. The budget still depends on items that are not quantified: fees and CIAC, receiving-system capacity, the grading plan, the geotech report, the detention decision and the building scope. Contingency should cover estimating uncertainty within the defined scope. The high-exposure items below should be closed with written quotes rather than absorbed by contingency.

Risk factor	Exposure	Finding / carry	Potential swing
Wastewater treatment & permitting	Moderate	Gravity to existing MH #2; no lift station/WWTP/septic. Receiving system and capacity unconfirmed. $0 treatment carried.	$0 to +$1.2M
Primary electric: CIAC & lead time	High	All primary/transformers by utility; no one-line. $3,000/site allowance. Transformer lead times can be 6–12+ months.	–$0.2M to +$0.5M
Geotechnical unknowns	Low–Moderate	No report referenced. Sands, deep water table likely; sewer up to 16 ft deep in caving sand.	$0 to +$150k
Stormwater & floodplain	Moderate	No detention shown; ties to existing 36" system. Bluff-edge 35–70% slopes.	$0 to +$400k
Entitlement & zoning	Low	Sealed IFB Rev 3; buffer/fence suggest approved site plan. Approval letter not in set.	$0 to +$100k
Impact & tap fees	High	No fee schedule; provider unknown. $3,000/site carried.	–$0.47M to +$0.9M
Phasing	Moderate	"PHASE I SITES" legend; boundary not legible. Shared items front-loaded.	See Phase 1
Earthwork (no grading plan)	Moderate	32,000 CY balanced assumed.	–$60k to +$250k

Phase 1

Sheet 06 has a "PHASE I SITES" legend, but the hatched boundary cannot be read in the version I analyzed. The workbook therefore uses a placeholder of 60 sites, with 60% of the linear network (roads, mains, sewer, storm, lighting) built in Phase 1. Front-loaded items go in at 100%: Div 01 general conditions, all tie-ins to existing water, sewer and storm, the 30–36" storm outlet, both electrical racks and panels, perimeter SESC, the fence, the buffer, parking, the garage and the pavilion. Per-site items are pro-rated. With those placeholders, Phase 1 is about $5.39M total, or about $89,800 per Phase 1 site, a 55% premium over the full-buildout $/site. Enter the actual Phase 1 lot count and linear share on the Inputs tab and the workbook recalculates.

Key Assumptions, Exclusions & Next Steps

- **Excluded:** homes, setup and foundations, garages, MH driveways, land, gas distribution, telecom/fiber relocation ("by others"), detention basin, off-site improvements, developer fee, and escalation beyond 2026 Q3.
- **Missing or illegible, replaced with assumptions:** grading plan, geotech report, electrical one-line and transformer schedule, E1/E2 lot assignment, part of the sanitary and storm lengths, structure tables, service schedules, Phase I boundary, pavilion size, sidewalk and parking extents, gross acreage, buffer planting schedule. The full list is on the Assumptions & Flags tab.
- **Next steps, in priority order:** (1) capacity and fee letters from the sewer and water provider; (2) a Consumers Energy CIAC and design quote; (3) the engineer's bid-form quantity sheet and grading model from Spicer Group; (4) geotechnical borings; (5) Drain Commissioner stormwater acceptance; (6) confirm the Phase I lot list.

Cost & Value Engineering Memo (full) [cost_ve]

MEMORANDUM

The Bluffs on Manistee Lake – North Expansion: Cost Estimate & Value-Engineering Review

To: Marc Maricondo, Jennings Realty Inc.

Date: September 28, 2026

Basis: Spicer Group plan set D-6631 (49 sheets), "03A – 16ft BOC Alternate – REV 3 – SEALED," issued for bidding 8/20/26. Owner: M Sites, LLC.

Companion: Bluffs_North_Expansion_Cost_Model.xlsx – live QTO, CBS, risk and VE model. Every number below comes from it; change the blue inputs to re-run.

Estimate class: AACE Class 3 (–10% / +20%). Quantities are from the PDF text layer; graphic-only items carry ranges.

Bottom line

- **155 new sites** (63 double-wide MH, 81 single-wide MH, 11 RV super sites) on ~26.1 ac of lots, ≈32 gross ac. Density is ≈4.8 sites per gross acre and ≈5.3 per net acre.
- **Base total project cost ≈ $8.5M** (range $6.8M–$11.2M): $6.95M hard including 12% contingency, plus $1.56M soft. That's **$54.9k per site** (range $43.8k–$71.9k), excluding land and homes. It sits at the low edge of the typical $55–90k ground-up MHC range because the expansion reuses the existing clubhouse and pool, uses 16′ one-way roads and utility-owned electric distribution, and needs no WWTP or lift station.
- **The largest cost risks aren't drawn:** where the sewer goes after existing MH #2 and whether it has capacity, water/sewer tap and capacity fees, and the absence of any geotech report. Carry them as separate owner allowances until resolved; together they could exceed the entire contingency.
- **Recommended plan: 165 sites, ≈$7.8M, $47.2k per site** ($52.6k like-for-like, excluding the driveway cost shift to home buyers). Stabilized NOI rises ≈$135k/yr (directional), taking yield on cost from ≈7.2% to ≈9.5%. Build it as a ~60-site Phase 1 (≈$3.9M).
- **Absorption, not construction cost, is the binding constraint.** A 55+ community that sells its own homes may absorb 15–30 homes a year, which means 5–10 years to fill 144 MH lots. Phasing and the RV/MH mix matter more than any single cost cut.

*__Caution on rents:__ **no comps or feasibility study were provided. Rents are escalated from a dated public profile of the existing community ($375–450/mo lot rent, $3,500 seasonal RV). Every NOI and value delta in this memo is directional.*

1. What the drawings show

System	Finding	Confidence / flag
Site count	63 DW + 81 SW MH lots + 11 RV super sites = 155. RV: all back-in (0 pull-through), all full hookup (water, sewer, 50/30/20A), concrete pad + patio. ADA-designated sites: 0. Also 15 existing lots get RV pads and/or drive approaches.	High. Phase I limits are graphic-only
Acreage & density	Lots 26.1 ac (avg MH lot 7,357 SF; DW 8,045 / SW 6,822 / RV 7,111 SF). Net (lots + roads) ≈29.2 ac. Gross not stated, estimated ≈32.1 ac (30.5–35).	Moderate / Low (gross)
Roads	≈6,463 LF of new private road. One-way (Minnesota, Montana, Pennsylvania, Washington) 2,604 LF at 16′ BOC. Two-way (California, Tennessee, Oregon, Kentucky) 3,859 LF. Section: 3″ HMA (1.5″ 5EL + 1.5″ 4EL) on 8″ 21AA with 7″ integral asphalt curb.	High / two-way width NOT shown – assumed 24′ BOC
Water	Extends the existing 6″ system (tapping sleeve at Kentucky 10+60, two plug removals). 7,609 LF 6″ C900 PVC, mostly in rear-lot easements. 48 valves, 10 hydrants, Mega-Lug + thrust blocks. At least 4 dead ends.	High. Source (municipal vs. well), fire flow and metering not shown
Wastewater	TREATMENT APPROACH: gravity collection only. ≈5,200 LF 8″ PVC at 0.40% min, ≈28 48″ MHs, 8–15′ deep (risers ≥14′), tying into EXISTING MH #2 (IE 647.56). No lift station, force main, package plant or septic in scope.	High for collection. Downstream treatment/capacity UNKNOWN – #1 diligence item

System	Finding	Confidence / flag
Stormwater	Enclosed HDPE: 12″ ≈1,050 LF, 15″ 333, 18″ 533, 24″ ≈755. ≈27 structures. Discharges to the existing 36″ trunk (CB #5018). No pond, detention or LID. Part 91 SESC (silt fence, stone access, inlet protection). NPDES construction NOC required (>1 ac).	Moderate. Some storm leads are mislabeled 'SAN'
Electrical	Utility provides primary, transformers and service conductors. Utility-direct meter pedestal per site: E1 100A, E2 200A, E3 100A loop-fed RV pedestal (14-50/TT-30/5-20 GFCI). Underground secondary. Two 100A common panels (NE, G). ≈20 25′ LED poles plus per-lot solar lamps.	Moderate. Transformer count/kVA not shown. Only 107 of 155 service tags legible
Gas	Not shown.	Allowance carried
Buildings	32′×40′ maintenance garage (1,280 SF; 'future' on E-47) and a pavilion (size not shown). No clubhouse, bathhouse, laundry, office, pool or dump station – the existing ones serve the expansion.	Moderate
Site amenities	930 LF privacy fence, 20′ Class A buffer (Alt. 3), MDOT signs (22), paved parking, sidewalk/trail, emergency drive, lighting. No gates, playground, dog park, storm shelter, mail kiosk or dumpster enclosures.	Moderate. USPS will require cluster mailboxes
Soils / regulatory	SSURGO: Covert and Plainfield sands, HSG A, non-hydric, including 67F 35–70% slopes at the bluff. No geotech referenced. No floodplain/wetlands shown (site ≈70+ ft above the lake). Setbacks: 12′ from road, 20′ home-to-home, 10′ to accessory, 25′ vision triangles. No conditions of approval printed.	High (SSURGO) / geotech missing

Full detail and the 16 drawing discrepancies and substituted assumptions are on the Findings and Flags tabs.

2. Quantity take-off (base, with ranges)

Category	Item	Base qty	Range / basis (confidence)
Earthwork	Clearing & grubbing	26.3 ac	21–30 ac; clearing limits graphic-only (Low)
	Topsoil strip (6″) / respread (3″)	24,900 CY / 96,300 SY	±20% (Low)
	Mass grading, balanced	37,100 CY	20,000–62,000; no grading plan (Low)
Paving	HMA 3″ roads / 21AA 8″ base	14,920 SY / 16,356 SY	13,700–16,500 SY HMA (Moderate)
	7″ integral asphalt curb	12,280 LF	11,350–13,400 (Moderate)
	Parking / emergency drive	900 SY / 350 SY	Graphic-only (Low)
	Concrete sidewalk 4″ / 8″ crossings	7,500 / 1,200 SF	4,000–15,000 SF (Low)
Site pads	MH driveways/aprons 5″	72,100 SF	Not dimensioned: DW 528 / SW 480 SF each (Low)
	Shed slabs 8′×10′ / RV pads 14′×60′ / RV patios	11,520 / 9,240 / 2,112 SF	Moderate
Water	6″ C900 main / valves / hydrants	7,609 LF / 48 / 10	High
	Services (1″ PE) / fittings	155 / 65	Moderate
Sanitary	8″ PVC gravity / 48″ MH	5,200 LF / 28	4,900–5,600 LF; 25–31 MH (Moderate)
	6″ services / risers ≥14′	155 / 20	Risers 10–35 (Low)
Storm	HDPE 12″ / 15″ / 18″ / 24″	1,050 / 333 / 533 / 755 LF	24″ range 438–800 (Moderate)
	Structures (48″ MH/CB / 24″ CB)	17 / 10	Moderate
Gas	Distribution	Not shown	$500/site allowance (Low)
Electrical	100A / 200A meter pedestals; RV pedestals	81 / 63; 11	Moderate
	Utility trench/conduit (developer)	7,000 LF	5,500–9,000 (Low)
	Transformers	By utility	Est. 18–25 × 50–100 kVA 1Ø (Low)
	25′ LED poles / lighting circuit / solar lamps	20 / 3,000 LF / 155	Moderate
Concrete	Total CY (walks, drives, slabs, pads, bldg)	≈1,970 CY	1,470–2,740 CY; no pool (Moderate)
Buildings	Maintenance garage (PEMB) / pavilion	1,280 SF / 1,200 SF	Pavilion 800–2,000 (Low)
Landscape/SESC	Seed & mulch / silt fence / inlet prot.	19.9 ac / 4,500 LF / 27	Plant counts not shown (Low)

3. Cost breakdown (CSI MasterFormat)

Pricing basis: 2026 US national-average installed unit costs × regional factor **0.90** (baseline RSMeans 30-city national average = 1.00; Northern Lower Michigan / Traverse City 496 region runs ≈0.85–0.93). Location is identifiable, so no national-average flag is needed, but confirm the factor with local bids. Utility-set fees are not factored. Unit costs for every line are on the CBS tab.

Division	Base	Low	High	Confidence
01 General Requirements (staking)	$58,500	$52,700	$67,300	Moderate
02 Existing Conditions (removals)	$22,500	$20,300	$25,900	Moderate
31 Earthwork & SESC	$661,100	$436,300	$1,023,000	Low
32 Exterior Improvements (paving, curb, walks, drives, fence, landscape)	$1,453,400	$1,050,300	$2,128,800	Moderate
03 Concrete (sheds, RV pads/patios, existing-lot pads)	$302,500	$237,400	$435,000	Moderate
10 Specialties (cluster mailboxes)	$27,500	$24,800	$38,700	Low
13 Amenity buildings ($/SF)	$206,100	$164,500	$290,900	Moderate

Division	Base	Low	High	Confidence
26 Electrical (incl. utility CIAC)	$937,000	$806,800	$1,151,800	Moderate
33 Utilities – water, sanitary, storm, gas	$2,103,400	$1,716,900	$2,551,000	Moderate–High
Direct hard cost	$5,772,100	$4,509,900	$7,712,400	
General conditions 6% + bonds/insurance 1.5%	$432,900	$338,200	$578,400	
Hard-cost contingency 12%	$744,600	$581,800	$994,900	
TOTAL HARD COST	**$6,949,600**	**$5,429,900**	**$9,285,700**	

Selected unit bases (national, pre-factor): 3″ HMA $19.50/SY; 8″ 21AA $13.50/SY; integral curb $8.50/LF; mass grading $4.25/CY; clearing $6,500/ac; 6″ C900 $52/LF; 6″ valve $2,900; hydrant $8,500; 8″ PVC sewer at 10–15′ $92/LF; 48″ MH $6,800; storm HDPE $52–92/LF by 12–24″; 100A/200A meter pedestal $1,950/$2,650; RV pedestal $3,100; 25′ LED pole $6,800; 5″ concrete drive $9.00/SF; PEMB garage $118/SF; pavilion $65/SF; utility CIAC $1,800/site (not factored).

Soft costs (itemized)

Item	Base	Basis (confidence)
Civil/electrical engineering & design	$310,300	5% of hard; sunk if paid (Moderate)
Construction admin, inspection, testing	$124,100	2% (Moderate)
Survey (lot stakes, as-builts, EGLE record dwgs)	$62,000	$400/site (Moderate)
Geotechnical + perc tests (none referenced)	$22,000	Required (Moderate)
Permits (EGLE Part 41, Act 399, Part 91, NPDES, LARA MHC, township, bldg/elec)	$45,000	Allowance (Low)
Water/sewer tap & capacity fees (not shown)	$387,500	$2,500/site; range $0–6,000 (Low)
Legal / entitlement / title	$65,000	Allowance (Low)
Owner's rep / CM	$155,100	2.5% (Moderate)
Builder's risk & GL	$24,800	0.4% (Moderate)
Construction interest carry	$233,000	65% LTC, 8%, 12 mo, 0.55 avg (Low)
Loan fees	$52,900	1% (Moderate)
Soft contingency	$74,100	5%
TOTAL SOFT COST	**$1,555,800**	**22% of hard**

Roll-up and benchmarks

Metric	This project	Benchmark / comment
Total project cost (excl. land, homes)	$8,505,400	Range $6.78M–$11.15M
Hard cost per site	$44,800	
Total cost per developed site	$54,900	MHC ground-up $55–90k; RV full-hookup $35–75k
Cost per gross acre	$264,600	MHC $300–600k/ac – below range (low density, shared amenities)
Cost per net developable acre	$291,300	

Benchmarks are general market knowledge (Midwest, 2025–26) and are not sourced comps; use them as a sanity check only.

4. Contingency and risk

Recommended hard-cost contingency: 12% ($745k); soft contingency 5%. A sealed bid set would normally justify 5–8%. The extra ≈4–7 points cover what this estimate can't measure: the two-way road section, parking/walk/drive extents, the missing grading plan, the missing geotech, and electrical tag gaps. Split it as ≈7% estimating/design completeness plus ≈5% construction change. Carry the three HIGH items below as separate owner allowances rather than folding them into contingency.

Risk	Rating	Why	Exposure
Wastewater approach & permitting	HIGH	Gravity to existing MH #2. The receiving lift station, treatment and capacity for +155 ERUs aren't identified. EGLE Part 41 required.	$0–750k
Primary electric CIAC & transformer lead times	MODERATE	Utility-owned distribution transfers procurement to the utility. The CIAC is unquoted and 1Ø pad-mount lead times of 20–50+ weeks could delay energization.	$100–400k + schedule
Geotechnical unknowns	HIGH	No geotech. Bluff slopes (35–70%), 12–15′ trenches in running sand, unverified pavement section and backfill reuse.	$75–350k

Risk	Rating	Why	Exposure
Stormwater / floodplain	MODERATE	No detention. Existing 36" trunk capacity unverified. HSG-A sand makes infiltration an easy fix. Floodplain risk low.	$0–300k
Entitlement status	LOW–MOD	Bid set is sealed. LARA MHC, EGLE and NPDES permits may be pending. Any density change reopens approvals.	$25–150k
Unshown impact/tap fees	HIGH	None shown. $2,500/site carried; municipal REU charges could run $4–6k/site.	–$0.39M to +$0.54M
Phasing / front-loaded shared infrastructure	HIGH	All sewer drains to one point, so the trunk, Oregon/California roads and water ties come first. A 60-site Phase 1 carries ≈50% of cost. Absorption is 5–10 years.	Carry cost (excluded)

5. Value-add and value-engineering alternatives

Deltas are vs. base. Negative hard Δ = capital saved. Value Δ = NOI Δ ÷ 6.25% cap. Net value = value Δ – (hard + soft Δ). Schedule effects and the regulatory/engineering constraint for each item are on the VE tab. Confidence is on the delta.

A. Cost-side VE (same product)

Alternative	Hard Δ	Net value	Conf.	Verdict / key constraint
A9 Driveways & shed slabs into home-sale package	–$672k	+$672k	Mod	Accept – a cost TRANSFER to buyers (+$5–6k/home), not a saving
A10 Defer maintenance garage & pavilion	–$206k	+$206k	High	Accept – deferral. E-47 already shows garage 'future'
A8 Delete per-lot solar post lamps	–$80k	+$79k	Mod	Accept – confirm R 125.1908 photometrics with poles only
A4 Valve rationalization (48 → ~30)	–$47k	+$44k	Mod	Accept – water-system owner standards
A6 Road base 8" → 6" 21AA on HSG-A sand	–$44k	+$42k	Low	Accept after geotech
A5 Native sand backfill, no import	–$32k	+$32k	Low	Accept after gradation tests
A1 Two-way road 24′ → 22′ BOC	–$27k	+$23k	Low	Only if the base is >22′. MHC 21′ minimum; fire access
A11 Concrete walk → 8′ asphalt trail	–$22k	+$20k	Low	If not a site-plan condition
A7 Terminal cleanouts vs. 4 MHs	–$19k	+$17k	Mod	EGLE Part 41; ends <150′ only
A3 Leaching basins / perforated storm	+$12k	–$30k	Low	PURSUE as hedge against a $0–300k detention requirement
A2 Delete curb, use swales	+$9k	–$24k	Mod	REJECT – a driveway every ~55′ needs 155 culverts. Plow damage. Hurts 55+ curb appeal
A12 Master-meter electric, private distribution	+$488k	–$528k	Mod	REJECT – base (utility-direct) is already optimal. Takes on transformer lead-time risk

Already optimal in the base, so no change is recommended: gravity sewer with no lift station (terrain allows), rear-lot water mains (they give Ten States separation without a common trench), 16′ one-way roads (already the 'BOC alternate'), and 100A 50/30/20 RV pedestals (right-sized for super sites). No over-specification of pipe class or pavement thickness is visible. The one exception is valve density (1 per ~160 LF).

B. Revenue-side value-add

Alternative	Capital Δ	NOI Δ/yr	Net value	Verdict / constraint (conf.)
B4 Water/sewer submetering (AMR)	+$75k	+$66k	+$989k	Accept IF the community pays a master water/sewer bill – unknown (Low)
B1 Density +10 sites (DW ~71′→65′, SW 55′→50′ frontages)	+$175k	+$41k	+$473k	Accept – avg lot stays ≈6,900 SF (>5,500 MHC avg). Needs replat plus LARA/EGLE resubmittal, +2–3 mo (Low–Mod)
B2 RV/boat storage on rear of deep lots 18–34 (~1.1 ac)	+$119k	+$28k	+$322k	Accept if access works. Lots stay ≥6,000 SF (Low)
B3 Premium pricing, ~20 oversized/corner/view sites	$0	+$11k	+$182k	Accept. Historic 'inflation-only' rent pledge may limit it (Low)
B6 Fiber/Wi-Fi conduit in common trench	+$24k	+$8k	+$110k	Accept. Cheap optionality; RV guests expect Wi-Fi (Low)
B7 Clubhouse fitness/craft addition	+$330k	~$0	–$330k	Defer to Phase 2 – supports absorption, not rent
B5 12 SW lots → RV super sites	+$74k	–$21k	–$411k	Stabilized NOI is lower, but RV fills in 1–2 seasons. Use only if MH absorption <20/yr
B8 Playground, splash pad, storm shelter, 2nd pool	—	0	—	REJECT – 55+ market; cost without rent

Amenities that move rent or occupancy here: the existing pool, pickleball, dog run and clubhouse (already built), garages (in the home package), Wi-Fi/broadband and RV/boat storage. Amenities that cost money without moving rent in this market: playgrounds, splash pads, a second pool, storm shelters (not required in MI). Resort-tier positioning is already largely achieved by the 60′×100′ concrete-pad super sites; the upside is in seasonal RV rate, not reclassification. Worth investigating, not scored: boat slips or a dock on Manistee Lake, if the parent parcel has usable frontage.

C. Phasing and capital

- **Phase it.** Minimum Phase 1 ≈60 sites at ≈$3.9M (≈$65k/site). Front-load the Oregon trunk sewer to EX MH #2, the water ties (Kentucky tap and California connections) plus one loop, and the Oregon/California/Kentucky roads. Defer the one-way street blocks, the north single-wide cluster, the garage, the pavilion and the clubhouse addition. The Phase I hatch on sheet 06 should be reconciled with this.
- **Utility ownership.** Electric is utility-owned (good: no transformer liability, no MPSC resale issues). Water/sewer extend an existing park system of unknown ownership and source. If it's private (well or lagoon/WWTP), it's a regulated operating liability, not a financeable asset, and capacity expansion becomes the gating capex. If it's municipal, it's neutral.
- **Optionality.** Loop 2–3 of the 4 dead-end 6″ mains (≈+$28k; better fire flow and water quality). Stub water, sewer and road ROW to any adjacent land under control (≈+$16k). The 8″ sewer at 0.40% has ample capacity (~0.5 MGD) for future growth, and nothing in the plan structurally forecloses expansion.

Recommended alternative plan

	Base	Recommended	Δ
Sites	155 (144 MH / 11 RV)	165 (154 MH / 11 RV)	+10
Total hard cost	$6.95M	$6.25M	−$0.70M
Total soft cost	$1.56M	$1.53M	−$0.02M
TOTAL PROJECT COST	**$8.51M**	**$7.78M**	**−$0.72M**
Cost per site	$54,900	$47,200	−$7,700
Cost per site, like-for-like (excl. A9 transfer)	$54,900	$52,600	−$2,300
Stabilized NOI (directional)	$609k	$743k	+$135k
Yield on cost (excl. land, lease-up)	7.2%	9.5%	+2.4 pts

Top 5 changes that get there:

- **B4 Water/sewer submetering** (+$75k capital, +$66k NOI) – confirm the park pays a master bill first.
- **A9 Driveways & shed slabs in the home package** (–$672k developer capital; price into homes).
- **B1 Density +10 sites** (+$175k, +$41k NOI) – one replat/resubmittal. Do it before bid award.
- **B2 RV/boat storage on the deep-lot remnant** (+$119k, +$28k NOI) – subject to access and zoning.
- **Spec rationalization bundle A4+A6+A7+A8+A11** (–$212k) – valves, 6″ base, terminal cleanouts, no per-lot lamps, asphalt trail. A5/A6 wait on geotech.

Also do, outside the totals: B3 premium pricing (no capital), C2 water loops, A3 infiltration pricing as an alternate, B6 conduit, A10 deferral, and the phased delivery above. Rejected: A2 (curb deletion), A12 (master-metering), B8 (non-rent amenities), and B5 unless absorption proves slow.

Immediate next steps

- Get a written capacity letter and fee schedule for the receiving water and sewer systems (or park-system O&M and capacity records).
- Order a geotech investigation: ~12–15 borings, gradations, perc tests, and a bluff-slope check.
- Get from Spicer: the two-way road section, the grading plan/earthwork balance, the Phase I limits, and CAD for a measured takeoff.
- File the electric utility load letter and line-extension application; request a CIAC quote and transformer schedule.

- Provide lot-rent and RV comps (or a feasibility study) so the Section 5 NOI deltas can move from directional to underwritten.

Sources

Plan set: Spicer Group D-6631-01 to -49, The Bluffs on Manistee Lake North Expansion (project file). Community background and historic rents: MHInsider, "Hybrid Approach Garners Success in Northern Michigan" (mhinsider.com/bluffs-on-manistee); MHVillage blog profile (blog.mhvillage.com/bluffs-on-manistee-lake); community website (thebluffsonmanisteelake.com); Michigan Manufactured Housing Association listing (michhome.org). Unit costs, regional factor and benchmarks are estimator judgment (2026 national average), not sourced bids.

Ideas and ROI (full, sourced) [revenue]

Bluffs North Expansion - Revenue & VE Within the Approved Plans Sep 28, 2026 · @Marc

Nineteen ideas pass the "could a reviewer find a discrepancy" test. If all the recommended ones are built, they add about $88k/yr NOI on $320k of incremental capital, which is about $1.39M of value ($1.07M net of cost). One item carries a large share of that: the water/sewer pass-through, at $40k NOI. It needs a legal review first. Without it, the total is about $48k NOI on $205k.

What the plan set shows

The Spicer Group set D-6631 (49 sheets, issued for bidding 8/20/26) is a civil and electrical set. It has no architectural drawings. Here is what it shows that bears on revenue:

- 155 new parcels (sheet 06): 63 double-wide lots (~80x100 ft, 7,600-11,700 sf), 81 single-wide lots (55x114 ft typical; lots 18-34 are 50.8x~173 ft), and 11 RV super sites (16, 17, 169-177). Sheet 06 also labels some sites "Phase I."
- Typical lot details (sheet 10): a 24x26 garage (double-wide) or 20x20 garage (single-wide), 12-ft setbacks, a concrete RV pad and patio on super sites, a 4-in slab for an 8x10 shed, and an Imperial III GS-97K solar lamp at each lot.
- Existing-lot updates: new concrete RV pads on lots 132-135 and 148-155, and drive approaches on lots 136-138.
- Shared elements: a pavilion (sheet 11 only), open space, a pedestrian trail and sidewalk, proposed paved parking, a 32x40 maintenance garage (marked "future" on sheet 47), 930 LF of privacy fence, a 20-ft Class A buffer with existing trees to remain, 25-ft vision triangles, and one-way 14-ft roads (the 16-ft back-of-curb alternate).
- Utilities: 6-in C900 water mains run along rear lot lines (sheets 33-42). Storm sewer is HDPE. Soils are Plainfield/Covert sands, hydrologic group A.
- Electrical: E1 100A and E2 200A home services. E3 pedestals are 100A loop-fed (Midwest U075CP6010: 50A, 30A, 20A GFCI). There are 25-ft SL1 80W LED poles on photocell/timer control. Panels NE and G sit on exterior racks, and Panel G is placed to feed the future garage.

Not on the plans: a pond, clubhouse, pool, courts, bathhouse, laundry, playground area, dumpsters, or an entry-sign area. The existing community already has a 5,000-sf clubhouse, a pool and hot tub, pickleball courts, and an

off-leash dog park (MHInsider, community site). Several amenity types you listed have no approved footprint in this set, so they are in the appendix, not the table.

Two counts to confirm with Spicer. The parcel tables give 144 home sites and 11 super sites. The E1/E2/E3 symbols I could read from the electrical sheets add up to about 101 home services and 7 pedestals. I worked from the text layer of the PDF, not the drawings. I used the parcel tables.

Assumptions

I couldn't get any rent roll or budget for the subject property, so every input below is an estimate or comes from a public source. Each one is flagged. The Excel model has each input in its own cell, and every ROI figure recalculates when you change one.

Input	Value	Basis	Grade
New home sites	144 (63 DW, 81 SW)	Sheet 06 parcel tables	Plans
New RV super sites	11 (parcels 16, 17, 169-177)	Sheet 06. The electrical sheets show about 7 E3 tags; see error register C7	Plans
Expansion base lot rent	$550/mo	ESTIMATE. The subject quoted $375-$450 "depending on home size and location" in an undated MHVillager piece, with inflation-indexed increases. UMH same-store is $585 (Q2 2026, Skyview). The earlier cost memo in this project used $525 SW / $575 DW	D
Stabilized MH occupancy	95%	ELS core 93.8%, Sun 97.8%, June 30, 2026 (Capright); the subject home side is "nearly full" (MHInsider)	B
Super-site seasonal rate	$6,995/season	Subject 2026 RV rate sheet. Seasonal range is $4,830-$6,995; super sites are $1,925/mo vs $1,325/mo for deluxe back-in. Replaces the $4,500 in the earlier cost memo	C
Share of the super-site premium from the add-on package	60%	ESTIMATE. The rest is the 60x100 pad, which is already drawn	D
Season	199 nights (mid-Apr to end of Oct)	Subject website	C
Transient full-hookup occupancy	68% of operating nights	OHI 2023 benchmarking, via MMCG 2026 outlook	B
Cap rate, MH-driven NOI	6.0%	UMH applied 6.0% to unencumbered NOI in Q2 2026; institutional trades are in the low-to-mid 4s (Skyview). A 55+ tertiary-market asset sits above that	B
Cap rate, RV-driven NOI	8.0%	Newmark 2026 survey, Class A/B RV going-in, via MMCG	B
Unit costs	See model	2026 northern Michigan installed-cost estimates. No quotes were obtained, so every cost is a range	D

Test applied to every idea: could a plan reviewer compare the finished site to sheets 01-49 and find a discrepancy? Anything that adds impervious area, new trenching outside approved runs, a new structure footprint, or a change of site type was moved to the appendix.

Part 1 - Revenue within the approved envelope

Premium site designation

The biggest source of returns here is pricing sites the plans already make different. It needs no capital. The subject already sets lot rent by "home size and location," so tiering the expansion follows the house practice.

Tier	Sites (sheet 06 parcel #)	Why it commands a premium	What to add inside the lot	Premium	
Premium	A	Oversized lots 35 (11,672 sf), 251 (11,526), 44 (10,796), 83 (9,473), 213 (9,204), 55 (8,672), 42 (8,595), plus about 5 lots with a Manistee Lake view, confirmed by drone at pad grade	30-50% more yard than the 8,000 sf DW standard; lake view	Sod front yard, a shade tree, and privacy screening at the side lines	+$50/mo
B	Deep SW lots 18-34 (50.8x~173 ft vs 55x114 typical); lots backing the 20-ft Class A buffer where existing trees remain; lots fronting the open space; the ~27 E2 lots with 200A service	Privacy, shade, rear-yard depth, and 200A capacity for a garage shop or EV	Same, at lower density	+$25/mo	
RV super	16 (10,645 sf) and 17 (8,742) are the largest super sites; 169-177 run 5,904-8,723 sf	16 and 17 can carry the top seasonal rate. On 169-177, back-in from one-way Washington Ave limits rig length (see error register P11)	The full super-site package (rows 6, 8, 11 and 12 in the table below)	Price 16 and 17 at the top of the rate sheet	
Base (no discount)	Lots next to the maintenance garage and emergency drive, the proposed parking, Panel NE/G racks, and the 3rd Street edge	Noise, headlights, equipment	Arborvitae screening on the lot side facing the nuisance	Base rent. Lease these last or use them for model homes	

Corner lots get no automatic premium. The 25-ft vision triangles (sheet 04) keep screening out of the corner, and a corner lot has two frontages.

Amenities and hardscape

- Already on the plans, evaluated as drawn: the pavilion (row 20), the concrete pads on existing lots 132-155 (row 7), and the super-site pad, patio and shed slab (the base for rows 6, 8, 11 and 12). The trail, sidewalk, SL1 lights, per-lot solar lamps, privacy fence and buffer are code or condition items. I didn't rank them as revenue amenities.
- Hardscape flag: no row adds impervious area. The patio upgrade is a finish on the drawn slab, and the shed sits on the drawn slab. Any new MH patio, walk, dog-run pad or parking pad would add area the storm design (sheets 11-21) didn't carry, so those are in the appendix. The one open risk is the pavilion's accessible route (error register P15). If the engineer adds a walk, it has to be counted against the approved impervious total.
- Existing clubhouse: FF&E there (row 9) touches no agency. It is outside this plan set but inside an existing approved building.

Ranked table - best to worst by cash-on-cash

These are 20 items (19 ideas plus the pavilion baseline). Costs are installed 2026 ranges; NOI and value use the midpoint. The Excel model holds every driver.

#	Amenity / idea	Installed cost (basis)	O&M /yr	Revenue mechanism	NOI /yr	CoC	Payback	Value @ cap	Grade and source	Permit	Conf.
1	Tier B pricing	$0	$0	+$25/mo on 40 Tier B lots at 95% occupancy	$11.4k	n/a	0	$190k @6%	D: inference from the subject's location-based rents	No permit contact	Mod
2	EV plug-in fee at RV pedestals	$0-300 (signage)	$0	$10/night on about 60 EV-nights where electric is included	$0.6k	400%	0.3	$7.5k @8%	D; KOA 2025 EV-ownership claim, methodology not disclosed	No permit contact	Low
3	Tier A pricing	$3-6k (drone view study + premium plat)	$0	+$50/mo on 12 Tier A lots	$6.8k	152%	0.7	$114k @6%	C: subject prices by location (MHVillager); waterfront RV +22.5% (Insider Perks via MMCG)	No permit contact	Mod
4	Metered RV pedestals	$5.2-8.9k ($400-700/pedestal incremental x 11, plus $0.8-1.2k reader/software)	$0.2k	Recovers seasonal electric of about $95/mo x 6.5 mo x 11 sites instead of absorbing it	$6.6k	94%	1.1	$82k @8%	C: the rate sheet includes electric only on daily/weekly stays. D: kWh estimate	No permit contact (covered by the project electrical permit)	Mod
5	Water/sewer pass-through with service meters	$94-137k ($650-950/site x 144)	$5.2k (reading and billing at $3/site/mo)	$55/mo per home, with a 50% haircut for rent pushback	$40.0k	35%	2.9	$666k @6%	D: no verified Manistee rate. Needs counsel review under Michigan MHC rules and a check against the subject's rent-guarantee marketing	Separate trade permit (plumbing)	Low
6	Super-site privacy screening	$11-17.6k ($1,000-1,600/site: 8 plants x 5-6 ft)	$1.1k	$350/season allocated share of the $2,165 super-vs-deluxe premium, on 11 sites	$2.8k	19%	5.2	$34k @8%	C: subject super sites include landscaping	No permit contact	Mod
7	Existing-lot RV pads, as drawn (132-135, 148-155)	$60-90k ($5,000-7,500 per pad x 12)	$0	+$8/night x 199 nights x 68% occupancy on 12 sites	$13.0k	17%	5.8	$162k @8%	B: OHI occupancy; D: $8 premium	On plans	Low
8	Gas fire table on the drawn patio	$9.9-17.6k ($900-1,600 each)	$1.1k	$300/season allocated, 11 sites	$2.2k	16%	6.2	$28k @8%	C: in the subject's super-site package	No permit contact (20-lb cylinder)	Mod
9	Clubhouse fitness FF&E	$18-35k	$1.5k	$3/mo retention and absorption allowance on 144 homes	$3.4k	13%	7.7	$57k @6%	None. No reliable source ties fitness to lot rent; reasoned estimate	No permit contact	Low

#	Amenity / idea	Installed cost (basis)	O&M /yr	Revenue mechanism	NOI /yr	CoC	Payback	Value @ cap	Grade and source	Permit	Conf.
10	Dog run fencing in the approved open space	$13-18k (~360 LF, 4-ft black vinyl chain link at $28-40/LF, plus double gate and waste stations)	$1.0k	$10/mo pet fee on 50 households, 50% attributed (the subject already has a dog park)	$2.0k	13%	7.8	$33k @6%	D	Separate permit (fence/zoning, if the township requires one)	Low
11	Super-site shed on the drawn 4-in slab	$27.5-41.8k ($2,500-3,800 each)	$0.6k	$400/season allocated, 11 sites	$3.9k	11%	9.0	$48k @8%	C: in the subject's super-site package	No permit contact (under 200 sf; verify with the township)	Mod
12	Stamped/colored finish on the drawn patio	$16-25k ($5-8/sf upcharge on ~288 sf)	$0.6k (resealing)	$250/season allocated, 11 sites	$2.2k	11%	9.4	$28k @8%	C: subject advertises stamped patios	No permit contact	Mod
13	Golf cart rental fleet (deferred)	$54-72k (6 carts x $9-12k)	$2.5k	$1,500/season-equivalent per cart	$6.5k	10%	9.7	$81k @8%	C: operator practice, no survey	No permit contact	Low
14	Sod on model, premium and super-site front yards	$22-36k (27 lots x 1,500 sf x $0.55-0.90)	$0	Avoids reseeding on Plainfield sand, about $110/lot. Absorption effect not counted	$3.0k	10%	9.9	$50k @6%	None for the absorption effect	No permit contact	Low
15	MH shed lease on drawn slabs (pilot)	$28-38k (10-unit pilot x $2,800-3,800)	$1.2k	$30/mo on 10 pilot units (most homes have garages)	$3.0k	9%	10.9	$50k @6%	D	No permit contact (verify)	Low
16	Pavilion FF&E and reservations	$4-7k	$0.2k	12 rentals x $50	$0.4k	7%	13.8	$7k @6%	None	No permit contact	Low
17	Shade tree on every MH lot	$58-94k ($400-650 per 2-in caliper tree)	$1.4k	+$5/mo, 60% realized while trees mature	$3.5k	5%	21.7	$58k @6%, below its cost	D	No permit contact	Low
18	EV Level 2 at proposed parking	$8-14k	$0.5k	150 sessions x $6	$0.4k	4%	27.5	$7k @6%	D. Residents charge at home, and the RV pedestals already have 14-50 outlets	Separate trade permit (electrical)	Low
19	Wi-Fi mesh on SL1 poles	$12-18k	$1.2k	$150/season on 11 super sites	$0.5k	3%	33	$6k @8%	A for demand: 48% of campers cite Wi-Fi (KOA 2025). D for the dollar premium	Separate trade permit (electrical) if added later	Low
20	Pavilion as drawn (baseline)	$55-95k (assumed 24x40 site-built; no detail in the set)	$1.0k	No measurable rent effect; the existing 5,000-sf clubhouse covers the gathering need	-$1.0k	-1%	never	-$17k	None	Separate building permit	Low

Most cost-effective amenity, and what to value-engineer

Best return per capital dollar: the metered RV pedestal. It returns 94% cash-on-cash and pays back in about 13 months, at moderate confidence. It beats the rest on this site for four reasons.

- The pedestal is being bought anyway. Only the $400-700 metered upcharge on 11 units is incremental, and it keeps the same Midwest 100A 50/30/20 receptacle set, enclosure and footprint.
- The rate sheet creates the revenue. The subject includes electric only on daily and weekly stays, and super sites book by the season (mid-April to October). An unmetered pedestal means the owner absorbs 6.5 months of air-conditioning load on every super site.
- The timing is free. Error register C7 already sends the RV feeder design back to the EOR. Metering can go into that same RFI response before the electrician orders equipment, so there is no rework and no plan change.
- The leading alternatives lean on weaker evidence. Tier pricing has zero capex, so it has no cash-on-cash ratio; it is pricing, not an amenity. The water pass-through needs a legal review.

The best-returning guest-facing amenity is super-site privacy screening, at 19%. The whole super-site package (screening, fire table, shed and stamped patio: rows 6, 8, 11, 12) costs about $7,600 per site and returns about 13% cash-on-cash, which clears the 8% RV cap rate. The subject's own ladder supports it: $1,925/mo for super sites vs $1,325/mo for deluxe back-in.

On-plan items that don't pay for themselves

On-plan item	Pays for itself?	VE move (same footprint, height and location)	Savings
Pavilion (sheet 11, "PR PAVILION")	No: -$1k/yr NOI on an assumed $55-95k	Specify a pre-engineered steel or glulam kit pavilion at the drawn plan dimensions, with a standard hip or gable roof, on a slab sized to the drawn footprint. The civil set has no pavilion detail (error register M7), so the specification is still open	$11-33k (20-35%)
930 LF privacy fence (sheet 04)	Not a revenue item. It is the Class A buffer condition	Only if the zoning approval doesn't name a material: 6-ft pressure-treated board-on-board instead of vinyl, at the same height and opacity. If the buffer spec names a material, leave it alone (error register P15)	$10-24k, conditional

Don't delete or relocate the pavilion. It may count toward the approved open-space or amenity package, and deleting it could change a condition. Deferring it past Phase 1 is fine only if no phasing or certificate-of-occupancy condition ties it to Phase 1.

The existing-lot RV pads (row 7, 17%) and the drawn super-site pad, patio and shed slab all pay for themselves. Leave them as drawn.

Part 5 - Implementation

The construction sequence follows the error register's pre-con agenda: sanitary, storm, water main, electric, curb and HMA, concrete flatwork, then landscape. Every item below stays inside what sheets 01-49 show.

#	Idea	What gets specified or installed, and where	Trade and when	Permit	Phase 1 or deferred; rough-in now	Cost and lead time	Operations after opening
1, 3	Tier A/B pricing	Premium plat of sheet 06 parcels. Drone view study at pad grade before homes are set	Owner/leasing, before lease-up and model placement	None	Phase 1. Nothing to rough in	$3-6k; 2 weeks	Premium lot rider in the lease; tier column in the rent roll; re-test premiums at 30% absorption
2	EV plug-in fee	Rule for NEMA 14-50 on the E3 pedestals; post it at check-in	Operations	None	Phase 1	Signage under $300	Line item in reservation software
4	Metered RV pedestals	Metered variant of the Midwest 100A loop-fed pedestal (same receptacles, NEMA 3R 16-ga enclosure, earth-burial mount), all 11 super sites. Put it in the C7 RFI response	Electrical, before equipment order (pre-con)	None beyond the project electrical permit; pedestal submittal	Phase 1. No rough-in needed	$5.2-8.9k; pedestal lead 4-8 weeks	Monthly reads (handheld or AMR), a tariff clause in the seasonal agreement, and a check that resale billing complies with Michigan PSC rules
5	Water/sewer pass-through	Meter yoke and pit with an AMR endpoint on each MH service stub. Written into the site-utility detail the EOR is already issuing (error register M5)	Utility contractor, during service installation behind the water main	Separate trade permit (plumbing); coordinate with the water provider	Phase 1 if counsel clears. Set the meter setting in the service detail now, even if billing starts later	$94-137k; meters 6-10 weeks	Billing vendor (~$3/site/mo), counsel opinion on Michigan MHC utility rules, and lease language for expansion residents only
6	Super-site screening	5-6 ft arborvitae or upright juniper on the side lot lines of parcels 16, 17, 169-177. Keep them 10 ft or more off the rear-lot water main (error register P13), out of 25-ft vision triangles, and clear of pedestal feeders (locate first)	Landscape, after final grade and electrical backfill	None	Phase 1	$11-17.6k; nursery stock 2-4 weeks	Watering the first season, plus a replacement warranty
7	Existing-lot RV pads	As drawn on 132-135 and 148-155	Concrete, with the flatwork package	On plans	Phase 1	$60-90k; ready-mix	Re-price these sites as deluxe on the rate sheet
8	Gas fire tables	Tabletop propane fire table (20-lb cylinder enclosed in the base) on each drawn super-site patio	Owner FF&E, at turnover	None (FF&E; no fixed tank)	Phase 1	$9.9-17.6k; 1-3 weeks	Cylinder exchange, winter storage in the maintenance garage, and a fire-safety rule in the site agreement
9	Clubhouse fitness FF&E	Light-commercial cardio and strength equipment in the planned fitness room of the existing clubhouse	Owner FF&E vendor	None	Time it to expansion lease-up	$18-35k; 4-8 weeks	Service contract (~$1.5k/yr), a waiver in the resident rules, and a check with the insurance carrier
10	Dog run	4-ft black vinyl chain link with a double-gate entry, waste stations and a bench, on turf in the sheet 04 "OPEN SPACE". No pad and no water line	Fence contractor, mobilized with the 930 LF privacy fence	Fence/zoning permit if the township requires one	Phase 1. Nothing to rough in	$13-18k; 2-4 weeks	Weekly turf and waste care; pet rider in the lease ($10/mo)

#	Idea	What gets specified or installed, and where	Trade and when	Permit	Phase 1 or deferred; rough-in now	Cost and lead time	Operations after opening
11	Super-site sheds	8x10 prefab wood shed, anchored to the drawn 4-in slab	Concrete pours the slab with flatwork; shed vendor delivers at turnover	None if under 200 sf; confirm with Manistee Township zoning	Phase 1. Pour every drawn shed slab now	$27.5-41.8k; 3-6 weeks	Keys and locks, and a roof and door check at each turnover
12	Stamped patio finish	Integral color plus a stamped pattern on the drawn super-site patio, same area and thickness	Concrete, at flatwork	None	Phase 1. Decide before the flatwork pour	$16-25k; no lead	Reseal every 3 years
13	Golf cart fleet	6 electric carts, stored and charged in the 32x40 maintenance garage	Owner, after the garage is built	None for carts. Charging uses garage circuits under that building's own permit	Deferred. Rough-in: Panel G is already placed to feed the future garage (sheet 49 note 15)	$54-72k; 2-6 weeks	Rental agreement, a liability rider, a battery maintenance vendor, and posted cart rules for the 7-ft one-way lanes
14	Sod	Sod instead of seed on about 1,500 sf per lot at the models, Tier A lots and super sites, on the same 3 in of topsoil	Landscape, at final stabilization (meets the 72-hr SESC rule)	None	Phase 1 for models; the rest as lots lease	$22-36k; days	Irrigate for 3 weeks
15	MH shed lease	Same 8x10 shed on drawn MH-lot slabs. Confirm the sheet 10 slab detail applies to MH lots, and keep sheds off the rear-lot main	Shed vendor, on resident demand	None if under 200 sf; verify	Pilot with 10 units	$2.8-3.8k/unit; 3-6 weeks	Shed lease addendum
16	Pavilion FF&E	Picnic tables and grills at the pavilion; a reservation calendar	Owner FF&E, after the pavilion is complete	None	With the pavilion	$4-7k	Reservation rules and cleaning
19	Wi-Fi (rough-in only)	Pull one unswitched 120V circuit (plus a #10 EGC per C8) through the approved 2-in SL1 conduit to the 4-6 poles nearest the super sites and pavilion. Mount APs later	Electrical, during the SL1 wire pull	Within the electrical permit now; separate electrical permit if added later	Rough-in now, build later	Wire ~$3-5k now; APs and backhaul $10-13k later	ISP contract and managed-Wi-Fi vendor
20	Pavilion (VE)	Pre-engineered kit at the drawn footprint and height	GC and kit supplier. Submit with the building permit	Separate building permit (needed regardless)	Per phasing conditions	Savings of $11-33k; kit lead 8-14 weeks	Annual inspection of roof fasteners

Not recommended: row 17 (shade tree on every lot) doesn't return its cost. Plant trees only on Tier A/B lots, where the rent premium carries them. Row 18 (EV Level 2) returns 4%. If demand shows up later, feed a charger from a spare breaker in Panel G or NE under its own electrical permit.

Part 6 - Summary

If everything recommended is built: about $88k/yr NOI on $320k of incremental capital, which is 27.6% cash-on-cash. That creates $1.39M of value at 6.0% (MH) and 8.0% (RV) caps, or $1.07M net of cost. The pavilion kit VE saves another $11-33k of capital. Without the water pass-through, the total is $48k NOI on $205k, worth $0.73M.

The totals exclude items already in the base bid (existing-lot RV pads, pavilion), deferred items (golf carts), rough-in-only items (Wi-Fi), and items that don't return their cost (EV Level 2, a tree on every lot).

Ideas ranked by NOI per dollar

Only 19 ideas pass the plan-discrepancy test with a traceable tie to these sheets. I didn't pad the list to 25.

1. Tier B lot pricing - $11.4k NOI, $0 capex
2. EV plug-in fee at RV pedestals - $0.6k, about $0
3. Tier A lot pricing - $6.8k, $4.5k
4. Metered RV pedestals - $6.6k, $7.1k

5. Water/sewer pass-through (after counsel review) - $40.0k, $115k
6. Super-site privacy screening - $2.8k, $14.3k
7. Existing-lot RV pads, as drawn - $13.0k, $75k (base bid)
8. Gas fire tables - $2.2k, $13.8k
9. Clubhouse fitness FF&E - $3.4k, $26.5k
10. Dog run in the open space - $2.0k, $15.5k
11. Super-site sheds - $3.9k, $34.7k
12. Stamped patio finish - $2.2k, $20.6k
13. Golf cart fleet (deferred) - $6.5k, $63k
14. Sod on models, Tier A lots and super sites - $3.0k, $29.4k
15. MH shed lease pilot - $3.0k, $33k
16. Pavilion FF&E - $0.4k, $5.5k
17. Pavilion kit VE - saves $11-33k of capital
18. Wi-Fi rough-in (conductor only) - about $4k now
19. Privacy fence material VE, if the buffer spec allows - saves $10-24k

Decide before the trade mobilizes

- Electrical (pre-con, with the C7/C8 RFI): metered pedestals; one unswitched circuit to the SL1 poles near the super sites and pavilion; spare breaker spaces in Panels NE and G
- Utility contractor (with the M5 site-service detail): a meter setting on every MH service stub
- Concrete (before flatwork): stamped and colored finish on the super-site patios; pour every drawn shed slab
- Fence contractor: dog run bundled with the 930 LF privacy fence mobilization; privacy fence material, if the buffer spec allows
- Landscape: sod areas; screening plant list and a locations plan that stays 10 ft or more off the rear-lot water mains and out of the vision triangles
- GC and building permit: pavilion kit selection, before the building-permit submittal
- Leasing: Tier A/B premium plat, before the first model home is set

Rough in now, build later

- Unswitched conductor in the approved 2-in SL1 conduit. This lets pole-mounted Wi-Fi APs or cameras go in later without trenching.
- Meter yokes on MH service stubs. Billing can start once counsel clears it, without digging up services.
- Shed slabs poured on every drawn location. Sheds get added as super sites and MH lots lease.
- Panel G on its rack, placed to feed the future garage (as drawn). That serves golf-cart charging once the garage is built.

Excluded - would require plan amendment

- Pickleball striping or a new court: no court or slab is on these sheets. The existing community already has courts.
- Playground: no approved play area. Equipment and fall-zone surfacing in the open space would be a new amenity footprint.

- Pool, bathhouse, laundry, new clubhouse, fitness building: no building footprint for any of them.
- Fishing or lake access, docks, kayak launch, bluff stairs: not on the plans. It touches 67F slopes and would need EGLE Part 301/91 review. No pond exists.
- Entry monument sign: no approved sign area. Sheet 04 shows MDOT regulatory and street-name signs only.
- Patios, walks or pads on MH lots, and any hardscape beyond the drawn super-site patio: adds impervious area the storm design didn't carry.
- Separate low-voltage or fiber conduit, or EV conduit to parking: new utility routing not shown. Only conductors inside the approved conduit are included.
- Retaining extra trees inside the approved clearing and grading limits: could conflict with the grading and SESC limits (sheet 05).
- Using MH sites as RV sites, or moving the home-to-RV mix: changes site type and count.
- +10 sites (earlier cost memo item B1): changes site count.
- RV/boat storage lot (earlier cost memo item B2) or storage on proposed parking: new use, and it could reduce the approved parking count.
- Water hydrant or spigot at the dog run: new water service off the approved mains.
- Deleting the pavilion, solar lamps, privacy fence or buffer plantings: each may be a condition of approval.
- Resident rentals inside the maintenance garage: a change of use of an approved accessory building. Golf-cart storage there for owner operations is kept.

Sources

All pages opened on 9/28/26.

- The Bluffs on Manistee Lake - RV Stays (2026 rate sheet)
- The Bluffs on Manistee Lake - Our Community
- MHInsider - Hybrid Approach Garners Success in Northern Michigan
- MHVillager - The Bluffs on Manistee Lake
- Michigan Manufactured Housing Association listing
- Capright - Manufactured Housing Market Update, September 2026
- Skyview Advisors - Q2 2026 Manufactured Housing Industry Report
- MMCG - US RV Park and Campground Market Outlook 2026
- NAI Outdoor Hospitality Brokers - KOA 2025 report summary
- PermitsGuide - Michigan shed permits (200-sf exemption)
- Dune Labs - Water submetering in Michigan
- Plan set: Spicer Group D-6631, sheets 01-49 (project file). Also this project's error register and cost memo from 9/28/26.

Two of these sources (Capright, Skyview Advisors) were spot-checked against their live pages 2026-09-28 and confirmed to state the exact figures cited here (Sun 97.8% / ELS 93.8% MH occupancy; UMH $585 same-store rent and 6.0% cap rate on unencumbered NOI). The rest were not independently re-checked.

Additional Ideas & Risks (deep research) [research_ideas]

Winter programming gives Bluffs North a real edge

The single clearest opportunity beyond the 19 ideas already captured is structural, not amenity-based: **no comparable property on Manistee Lake — including North's Edge RV Resort & Marina and the existing Bluffs property itself — operates true four-season revenue**, running instead a compressed April/May–October/November season and closing for winter (North's Edge RV Resort & Marina; The Bluffs on Manistee Lake). That gap is real money left on the table given Manistee County's DNR-mapped snowmobile-trail network, working winter-camping precedents elsewhere in Michigan, and marina-slip economics in comparable Midwest lake markets running **$1,600–$7,000 per slip per season** (Midwest Waterfront Marina, RV & MH Community listing, LoopNet). Beyond the winter play, the strongest national ROI data points toward solar-plus-shade structures (documented rent premiums and vacancy cures at comparable RV resorts) and toward treating the existing Bluffs first phase — not generic Sunbelt benchmarks — as the primary absorption anchor for underwriting the 155-site expansion, since that phase absorbed **roughly 49 of 51 manufactured-home sites over about six years** (MHInsider). On the cost and financing side, Manistee County's sandy, well-drained soils specifically favor trenchless utility installation and infiltration-basin stormwater management over conventional trenching and detention ponds, and Michigan-specific USDA and C-PACE programs exist but each has a real eligibility catch that needs to be worked through before counting on the money. The largest open risk is regulatory rather than financial or amenity-based: Michigan's High Risk Erosion Area program and Manistee Township's own bluff-setback ordinance are both written specifically around the Lake Michigan shoreline, and it is currently unconfirmed whether either applies to this bluff above Manistee Lake — a single parcel-level lookup that could materially change the buildable envelope near the bluff edge.

Marina slips, snowmobile access, and heated storage turn winter from a closure into a season

Every Manistee Lake comparable found in this research runs the same compressed season and none has adapted with real winter programming, which means a genuine four-season strategy would differentiate the expansion from the entire local competitive set rather than merely catch it up. **North's Edge RV Resort & Marina**, the closest direct comp on the same lake, operates **146 marina slips** (bow-in or stern-in, with 30-amp power and city water at each slip) plus a covered stainless-steel fish-cleaning station and 57 full-hookup RV sites, and partners with **RVshare** for peer-to-peer motorhome rentals — but its own season runs **April 1 through November 1 only** (North's Edge RV Resort & Marina). The Bluffs' own current site content lists a pool, hot tub, pickleball, and clubhouse, with fishing and boating framed as nearby rather than on-site, and shows no marina, storage program, or winter messaging at all (The Bluffs on Manistee Lake — Our Community). Manistee County already has the infrastructure a winter product would ride on: a DNR-mapped, actively maintained **designated snowmobile trail system**, funded in part through the Michigan DNR's Snowmobile Trail Improvement Fund, with local Benzie–Manistee clubs coordinating development (Michigan DNR Snowmobile Trail Maps — Manistee County; Michigan DNR Snowmobile Trail Improvement Fund). Other snowbelt operators prove the model works commercially: Along the River in New Hampshire markets "direct snowmobile trail access from RV sites and cabins to 7,000 miles of trails" and runs winter as its own distinct camping season rather than an off-season, and Michigan's own **Lake Chemung Outdoor Resort** runs winter RV camping with dedicated snow removal, heavy-duty electrical service for space heaters, and **ice fishing inside the park itself** as a marketed amenity, not merely a nearby attraction (Along the River NH; Lake Chemung Outdoor Resort). A portable pricing mechanic comes from an Arizona comp offering winter-monthly full-hookup sites at **$500/month plus electric**, contingent on committing through the full October–April season — a structure that discourages partial-season churn and could transfer directly to a Manistee winter-monthly product (4 Seasons RV Park Arizona — Rates).

The marina piece has real, evidenced economics from a genuinely comparable Midwest freshwater-lake market: a Chain O'Lakes-area (Antioch, IL) property combining **100 wet slips with 45 seasonal trailer sites** projects roughly **$225,000/year from trailer sites and $88,000/year from standalone slips**, with recent per-slip sale

values of **$12,000–$15,000** and standalone seasonal slip rental around **$1,600/season** (trailer sites with a slip included running **$3,775–$7,000/season**) (Midwest Waterfront Marina, RV & MH Community, Antioch IL — LoopNet). That range suggests even a modest slip count at a Manistee Lake marina could plausibly add low-six-figures in annual revenue, though this is a market analog and not a site-specific pro forma. Indoor/heated RV and boat storage is a second, stackable winter revenue line: national peak demand for such storage runs October–March, heated storage typically commands **$100–$450/month**, and tiered pricing (outdoor → covered → heated) is explicitly recommended to capture a wider range of renters, including an affluent tier willing to pay for climate control (Storable — 7 Ways to Boost the Value of Your RV and Boat Storage Facility; Toy Storage Nation — Maximizing Revenue in RV/Boat Storage). Boat shrink-wrap and winterization is already an established freestanding service industry across Michigan lake country, evidenced by multiple operating providers in comparable lake markets (Rapid City, Interlochen, Grand Rapids), which suggests a bundled on-site winter-storage-plus-winterization program is a realistic add-on even though no source quantified per-boat revenue for a resort-operated (rather than third-party) version of that service (Luhrs Landing, Rapid City MI; Long Lake Marina, Interlochen MI). None of this is proven at the property level — no source gave concrete winter-occupancy or revenue figures specific to northern Michigan or Manistee Lake, and it remains unconfirmed whether The Bluffs has any undisclosed marina plans — but the absence of any four-season competitor on this specific lake is itself the strongest evidence that this is where the expansion could out-position the local market rather than just match it.

National amenity data favors solar-plus-shade and screens out glamping for this demographic

Beyond the winter opportunity, 2025–2026 industry data points most strongly toward **EV charging, solar-with-shade combinations, and glamping/hybrid accommodations** as the clearest quantified-ROI amenities, while 55+-specific sources emphasize athletic courts and social programming as the strongest draw for this exact resident base — a useful filter for screening which national trends actually fit a 55+ MH/RV community rather than a general-population RV park. Sector fundamentals are strong to begin with: 2025 manufactured-home occupancy averaged **95% overall and 97% in 55+ communities specifically**, with site rent up **5.9% in 55+ communities** for the year, and MHVillage specifically flags rising demand for **pickleball, tennis, and shuffleboard** in retirement-oriented communities — directly reinforcing (rather than replacing) the already-planned clubhouse-fitness item (MHInsider — Manufactured Housing Industry Trends & Statistics; MHVillage — Top Community Amenities). Solar-plus-shade has the most concrete, multi-source ROI evidence of any amenity researched: **Auburn Gold Country RV Park** generates **$90,000/year in electrical value at over 20% annual ROI**, the **Tucson/Lazydays KOA** avoids roughly **$330,000/year** in electricity costs across three installations while its shaded sites earn about **$3,800/year more per site** than unshaded ones, and **30 previously vacant sites became revenue-generating** once shade/solar canopies were installed — a direct, cited vacancy-reduction effect, not just a rent premium (Woodall's Campground Magazine — Park Owners Realize Benefits of Utilizing Solar Power). Payback periods across cited installations run **3–8 years**, and Bellefonte/State College KOA separately earns **~$8,000/year** selling green energy credits on top of the direct savings — though every quantified example found is in a warmer, higher-sun climate, so northern Michigan's shorter winter daylight window likely changes the generation math in ways this research could not independently verify.

Two other trends are worth flagging precisely because they don't clearly fit this property. **Glamping accommodations** command a **2–5x nightly-rate premium** ($150–$300/night versus $30–$100 for standard RV sites) and are scaling fast industry-wide — roughly 465 glamping units added across 23 parks by 2026 — but glamping's target market of younger experience-seekers likely doesn't overlap well with a 55+ age-restricted resident base, making this a stronger fit only if the expansion carries a meaningful transient/guest RV mix rather than being purely long-term residency (Innowave Studio — Trends in the U.S. RV Park and Resort Sector 2025-2026). **Battery backup** is similarly not yet a standard amenity even among large, well-capitalized operators: **Sun Communities** is still in a feasibility-study phase for a combined solar-plus-battery-plus-natural-gas "Nex-Gen" system explicitly framed around hurricane resilience in Sunbelt states, meaning any Michigan pitch for battery

backup would need to be built on a winter-storm-outage rationale rather than the hurricane logic actually cited in the source material (Wikipedia — Sun Communities, citing company disclosures). EV charging is a lower-cost, more straightforwardly portable addition — **40% of EV-owning campers** say they're more likely to choose a park with charging, and **48% of campers** rate Wi-Fi as "very important," reinforcing the already-planned Wi-Fi rough-in (Innowave Studio) — though it skews more toward attracting RV transients than 55+ permanent residents specifically.

The property's own six-year lease-up is a better underwriting anchor than any outside comp

Because no Midwest- or Great-Lakes-specific absorption benchmark exists in trade publications, the single most useful data point for underwriting the 155-site expansion is the Bluffs' own first-phase history, not a national or Sunbelt-derived assumption. The Bluffs on Manistee Lake was developed by James R. Dougherty of Community Management Group, opened and began selling homes in **late fall 2018**, with RV spaces opening to vacationers in **spring 2019**; the first phase built out to **51 manufactured-home sites and 44 RV spaces**, with **150 additional sites already engineered** for expansion — nearly matching the 155-site plan now under consideration (MHInsider, "Hybrid Approach Garners Success in Northern Michigan"). As of the MHInsider interview, the property was "nearly full on the home side," implying roughly **49 of 51 sites absorbed over about six years**, with COVID-era demand specifically noted as having delayed home sales while increasing RV interest — an uneven, not linear, absorption path (MHInsider). Monthly site rent runs **$375–$450** with a marketed "lifelong guarantee that will only rise with the rate of inflation," the RV seasonal rate is **$3,500**, and a base "Lake House" home model prices at **$116,000** ($155,000 fully equipped with garage); occupancy splits roughly **50% year-round and 50% seasonal**, with seasonal residents wintering in Florida, Arizona, and Texas — a reverse-snowbird pattern distinct from a Sunbelt park's winter-peak model (MHVillage blog, "The Bluffs on Manistee Lake"). The same source documents a real, named conversion dynamic worth building into the second phase's marketing: seasonal RV guests function as "a ready-made pipeline for community residence," i.e., a proven renter-to-buyer path already operating at this exact property. National scarcity context underscores why this in-market data point matters more than usual — only **315 new manufactured-home communities** were built nationwide over a recent 16-year span, versus **2,645** in the prior 15 years, making a true expansion like this a rare, hard-to-benchmark event industry-wide (MHVillage blog).

Other Michigan comparables round out the picture but mostly confirm that cost and financing terms are not publicly disclosed anywhere in this sector. **Dancing Fire Glamping & RV Resort** in Port Huron Township reopened July 25, 2025 after an 18-month renovation to **140 RV sites** with a new clubhouse, pickleball court, pool, cabins, and event pavilion, but disclosed no cost, financing, or absorption figures (RVBusiness). **UMH Properties** (public MH REIT) closed on **Hidden Creek** in Erie, Michigan in July 2022 — 351 developed homesites at roughly **63% occupancy** at acquisition, though its post-acquisition lease-up trend was not found in this research pass (UMH Properties press release). **Sun Communities**, headquartered in Southfield, Michigan, is the dominant public operator in-state with **92 Michigan communities (≈16.6% of its national portfolio)** as of year-end 2024, and has disclosed portfolio-wide (not Michigan-isolated) expansion-site totals in the **1,000–2,200 sites/year** range across several recent years — confirming that lenders and large operators in this space do routinely underwrite pad-count expansions, even though no Michigan-specific rate or absorption terms were found in the sources reviewed (Sun Communities Q4/FY2024 results; RVBusiness). No Michigan presence was confirmed for Equity LifeStyle Properties in this pass, and Midwest rent growth was the steepest of any U.S. region in 2025 at **7.2%**, reinforcing that this submarket is supply-constrained rather than slow to absorb (NorthMarq).

Sandy soil and Michigan-specific programs offer real, if narrow, savings

Manistee County's sandy, well-drained (HSG A) soils specifically favor two cost-reduction levers beyond the existing value-engineering list. **Trenchless/horizontal directional drilling (HDD)** for water and sewer lines can cut total installed cost — including surface restoration — by roughly **30–50%**, with one documented case showing a

40% reduction once pavement replacement and traffic control were included; sandy soil is explicitly favorable for this method because "drilling through soft soil like sand is faster and cheaper" than clay or rock, and HDPE pipe commonly paired with HDD carries a **50–100 year** functional lifespan (Timothy Rose Contracting; Underground Drilling LLC). Because the site's soil already has a high natural infiltration rate, stormwater management can likely shift from a conventional detention pond to **infiltration basins**, which are simple and inexpensive to build (**$0.50–$1.30 per cubic foot**) and recharge groundwater rather than holding a permanent pool — though they carry "one of the highest failure rates of all stormwater practices, almost always due to a lack of maintenance," so any adoption needs a real maintenance commitment built into the operating budget, and Michigan-specific EGLE/MDEGLE approval for this approach was not independently confirmed (Stormwater Center — Infiltration Basin fact sheet). A lower-volume-road pavement swap — chip seal instead of full-depth hot-mix asphalt — is documented as up to **40% cheaper** on low-traffic roads with weak subgrade, though natural sand aggregate needs care given skid-resistance concerns, and this may already partly overlap with the existing "thinner road base" item (Geotechnics journal).

On the incentive side, most Michigan-specific programs are a partial fit rather than a clean match, and each needs a direct confirming call before being counted on. **USDA's Rural Economic Development Loan & Grant Program (REDLG)** offers genuinely zero-interest capital (the local electric co-op may add up to 1% administrative interest) over 10-year terms, but it is channeled through rural electric cooperatives — and Manistee County's likely utility, Consumers Energy, is an investor-owned utility rather than a cooperative, so eligibility needs direct confirmation (Piedmont Electric, REDLG Q&A). **USDA Section 524 site loans** directly match "acquire land, install utilities and roads for finished lots" at a below-market fixed rate, but are restricted to non-profit or public-entity applicants, a likely disqualifier for a private for-profit operator absent a workaround structure (USDA Rural Development, Rural Housing Site Loans in Michigan). **C-PACE financing via Lean & Green Michigan** can cover up to 100% of hard and soft costs with no upfront cost, repaid over up to 25 years through a special assessment — a genuinely attractive, low-friction tool, but scoped only to the energy- and water-efficiency-qualifying slice of the project (efficient electric infrastructure, water-conservation systems, on-site renewable generation), not general road, water-main, or sewer-trunk construction (Lean & Green MI — How PACE Works). Manistee County has **no designated Opportunity Zone**, so that lever is unavailable regardless of structure (OpportunityDb, Manistee County listing).

For the broader capital stack, stabilized manufactured-housing communities can access agency-style (Fannie Mae/Freddie Mac product-set) permanent financing at up to **80% LTV** and a minimum **1.25x DSCR** over 5–15-year terms once a community has at least 50 pads and professional management, with 2026 market rates for strong stabilized parks in the **7.5%–9.5%** range versus **10%–13%+** for transitional or bridge financing (Commercial Loan Direct; Real Estate Solutionist, MHC Financing & Stabilization Guide 2025). Because the expansion adds onto an already-operating, presumably stabilized community, financing it against the whole community's blended cash flow — rather than pricing it as a standalone transitional deal — could plausibly land in the cheaper stabilized-rate range, though this is a structuring question for whoever leads financing, not a confirmed outcome. **SBA 504 loans** explicitly cover land purchase and site-development costs (grading, streets, utilities) through a 50/40/10 structure, but require the property to be more than 51% owner-occupied by the borrowing business — a real constraint for a pure real-estate-holding entity that needs direct confirmation from an SBA-approved lender. Finally, **Consumers Energy** markets an "innovative economic development rate" and staffs a dedicated Economic Development team with named regional contacts (Crystal Stiles, Senior Executive Director of Growth; Lisa Pung, Site Readiness Program Manager) — a real, likely-productive lever for offsetting the electric-extension CIAC cost, but its actual rate formula and eligibility criteria are not publicly indexed and require direct outreach (Consumers Energy, Economic Growth; Consumers Energy, Company Economic Development page).

An unresolved question about which shoreline rules apply is the top regulatory risk

The most actionable, lowest-cost, highest-value item in this entire research pass is a single unresolved regulatory question: whether Michigan's **High Risk Erosion Area (HREA)** program and Manistee Township's own bluff-setback ordinance — both written specifically around the **Lake Michigan** shoreline — actually extend to this bluff above **Manistee Lake**. Manistee Township parcels have been designated under HREA since **December 20, 1993**, and any structure anywhere on a designated parcel (not just near the shoreline) triggers a permit requirement and a **60-year Projected Recession Distance setback** for non-movable structures (30 years for smaller movable ones) (EGLE — High Risk Erosion Areas: Program and Maps). But the Township's own zoning ordinance defines "Bluffline" explicitly as the crest "above the beach adjoining **Lake Michigan**," and the parallel state HREA program is a Part 323/Great Lakes-coastal-zone-specific statute — meaning it is genuinely unconfirmed whether either rule applies as written to a Manistee Lake-facing bluff, even though Manistee Lake connects to Lake Michigan via the Manistee River channel (Manistee Township Zoning Ordinance). If the parcel is not HREA-designated and the Lake-Michigan-specific bluff-line rule doesn't apply, the project avoids the 60-year setback entirely and the bluff constraint becomes a purely geotechnical and ordinary soil-erosion-control matter; if it does apply, that setback becomes a hard limit on the buildable envelope near the bluff edge. The source PDF listing Manistee Township's designated HREA parcels returned an access error during this research and should be opened directly, or checked against EGLE's public HREA GIS layer, before site planning proceeds.

On permitting mechanics, the news is more favorable than a worst-case read of Michigan's 150-day statutory review clock for sanitary sewer construction (Part 41) would suggest. Because the expansion ties into the existing MH #2 system rather than building new treatment infrastructure, it likely qualifies for **EGLE's expedited Part 41 review track**, available to gravity sewer extensions of **10,000 feet or less** (or duplex pump stations with **2,000 feet or less** of force main) connecting to an existing collection system — filing a notice of intent **10–20 business days** ahead of the full application is the concrete next step (EGLE — Expedited Application Review Information). The real gatekeeper on capacity, though, isn't EGLE at all — it's whoever owns and engineered the existing MH #2 system, and a written capacity-allocation letter for **155 additional connections** should be secured before spending money on formal design. Soil erosion control under **Part 91** is triggered here on two independent grounds (any disturbance over 1 acre, and any earth change within 500 feet of a lake or stream) and is administered locally by the **Manistee County Planning Department**, not EGLE directly, with the state's NPDES construction stormwater permit (MIR100000, a five-year general permit) riding on top of that local approval as a Notice of Coverage rather than a separate lengthy state review (Manistee County Soil Erosion Permit Application; EGLE — Soil Erosion and Construction Storm Water). The Manistee County Drain Commissioner separately reviews site stormwater and detention design. Geotechnically, Michigan's Great Lakes-region bluffs are typically composed of glacial till and clay prone to rapid wave erosion and freeze-thaw destabilization, and a proper bluff-stability investigation — survey, soil borings, slope-stability analysis, drainage assessment, and coastal wave analysis if lake-facing — is a well-established, routinely performed scope at firms with Great Lakes shoreline experience such as GZA; general (non-bluff) geotechnical cost benchmarks of **$1,500–$5,000** for slope-stability analysis and **$300–$900 per boring** should be treated as a floor, not a project estimate, for a 155-site development on a mapped 26–70% slope (GZA — Shoreline Protection and Bluff Stabilization, Lake Michigan; Angi — Geotechnical Report Cost).

Conclusion

The pattern across all four research threads is the same: the most valuable moves available to Jay are the ones no comparable Great Lakes property has made, not the ones every Sunbelt RV resort already has. Winter programming — a marina, snowmobile-trail-linked marketing, on-site ice fishing, and a winter-monthly storage product — is differentiated precisely because it's absent from every Manistee Lake comp studied, including the sibling Bluffs property itself, while solar-plus-shade is the one national amenity trend with hard, multi-site evidence

of both a rent premium and a vacancy cure. The financing and permitting pictures reward the same instinct to use what's already on the ground: the existing Bluffs first phase is a far better absorption anchor than any Sunbelt benchmark, the existing MH #2 sewer system is the fastest path through EGLE's expedited review, and the site's own sandy soil is what makes trenchless utilities and infiltration-basin stormwater genuinely cheaper rather than merely theoretical. The one item that could reorder every other priority is the Manistee Lake HREA/bluff-line question — a fact this research could not resolve but that a single parcel-level GIS check or a call to EGLE Shoreland Management and the Township zoning administrator can settle quickly, and which should be resolved before the buildable envelope near the bluff edge is finalized in site planning.

Long-Term Hold -- Class 4 Estimate Rev 1 (full, 10/2/26) [long_term_hold_class4]

Preliminary Cost Estimate — AACE Class 4 — Rev 1: Long-Term Hold Specification

The Bluffs on Manistee Lake — North Expansion | Manistee Township, Manistee County, Michigan

Owner: M Sites, LLC | Engineer: Spicer Group | Plan set D-6631 (49 sheets), 16-ft BOC Alt., IFB 8/20/26 | Rev 0: 9/28/2026 → Rev 1: 10/2/2026 | Pricing 2026 Q3

Companion workbook: Bluffs_North_Expansion_Class4_Estimate_Rev1_LongTermHold.xlsx (live formulas; it governs). Preliminary and non-binding. Rev 1 supersedes Rev 0, and replaces the conflicting recommendations in the 9/28 Cost & VE Memo (see Section 6).

Executive Summary

Owner direction (10/2/26): M Sites is a permanent owner. The community will never be sold. Every specification is now chosen for the lowest 40-year cost of ownership, not the lowest first cost. In practice that means three rules:

- **Rule 1.** Anything buried or structural is built right the first time: road base, pipe class, valves, conduit, curb, electrical service size. None of these can be upgraded later without tearing up a finished, occupied community.
- **Rule 2.** Delete components that create a perpetual replacement cycle, such as per-lot solar lamps and painted markings.
- **Rule 3.** Keep what is already the right long-term answer: utility-owned electric, gravity sewer with no lift station, and rear-lot water mains.

Metric	Rev 0 (9/28)	Rev 1 (Long-Term Hold)
Total direct hard cost	$6,503,858	$8,638,002
Hard cost incl. 12% contingency	$7,284,321	$9,674,562
Soft costs incl. financing + 10% contingency	$1,687,112	$2,192,080
TOTAL PROJECT COST	**$8,971,433**	**$11,866,642**
Cost per site (155)	$57,880	$76,559
Like-for-like per site (excl. owner-installed driveways)	$57,880	$70,969
Class 4 range	$7.18M – $11.66M	$9.49M – $15.43M
Phase 1 (placeholder 60 sites)	$5.39M	$7.13M ($118,836 / site)
Recommended capital reserve	not addressed	$216 / site / yr ($18 / mo)

Where the +$2.90M goes. About $0.87M all-in is MH driveways now in developer scope. They are owner-installed to one community spec, and the cost is recoverable in the home price. About $0.11M is a correction: Rev 0 understated the two-way roads by pricing them at the 16-ft one-way width. They are now 24-ft back of curb, per Plan Review item C9. About $0.13M fixes errors flagged in the plan review: RV power source, grounding-conductor size, hydrants, storm upsizing, and utility stubs to the garage and pavilion. The remaining roughly $1.8M is long-

term upgrades plus the general-requirements, contingency and soft-cost markups on them. The largest upgrades are concrete curb and gutter, 200A service on every MH lot, 4" asphalt, telecom conduit, AMR-ready water meter pits, SDR-26 sewer and a heated maintenance shop.

Does it pay? I modeled nine systems over 40 years at a 4% real discount rate. Together they carry $0.50M of extra capex and avoid $0.74M (present value) of replacement and operating cost, for a net +$0.24M. Two items do not pay back on present value alone, and I show that openly in Section 4: concrete curb and gutter, and the water-main loops. I still recommend both, for drainage function and fire-flow resilience. Several other items cannot be quantified but matter more over a 40-year hold: conduit, base thickness, tracer wire, lot-line cleanouts and GIS as-builts.

Benchmark. At $76.6k per site, the project is within the ground-up MHP range ($55k–$95k) and above the typical expansion range ($40k–$70k). That is expected. Benchmarks reflect first-cost-driven builds and exclude driveways. On a like-for-like basis it is $71.0k per site, right at the top of the expansion range, with a materially lower long-term cost of ownership.

1. Principles Applied

Principle	What it changed
Buried and structural work is done once	8" base kept (VE A6 reversed); SDR-26 sewer; all valves kept; tracer wire; conduit and road-crossing sleeves; 200A service on every MH lot
Delete perpetual replacement cycles	Per-lot solar lamps replaced by 10 hard-wired aluminum poles; paint replaced by thermoplastic; wood fence replaced by vinyl/composite
The owner controls everything on its land	MH driveways built to one community spec by the owner's contractor (cost recovered in home price); maintenance shop built in Phase 1
Drainage that works on day 10,000	Concrete gutter gives a true flowline (fixes the C3 ponding risk); 12" pipe at 0.26% upsized to 15"; infiltration basins at sags
Know what you own	Expanded geotech, sealed grading plan, digital CCTV baseline of sewers, survey-grade GIS as-builts, full-time owner inspector, 2-year warranty bond
Size trunk infrastructure for full buildout	Phase 1 builds trunk sewer, water ties and loops at final size; never undersize to save Phase 1 cash

2. Rev 1 Specification Changes

Costs below are regional-adjusted direct hard cost, before markups. Line-by-line detail is on the "Rev1 Spec Changes" tab.

System	Rev 0 spec	Rev 1 long-term spec	Δ capex
Road surface	3" HMA, 16-ft width on all roads	4" HMA (2.5" + 1.5"); top course placed after homes are set; two-way roads corrected to 24-ft BOC	+$176k
Curb	7" integral asphalt curb	Concrete mountable curb & gutter (fits the EJIW 7045 frames already specified)	+$196k
Road base	8" 21AA (VE memo proposed 6")	8" minimum, confirmed by geotech; 1,500 CY undercut allowance	+$72k
MH driveways	Excluded (home-sale scope)	5" concrete, owner-installed to community spec, recoverable	+$637k
Water	7,609 LF 6", 38 valves, 10 hydrants, 4 dead ends	+600 LF loops, 41 valves, 14 hydrants, tracer wire, AMR-ready meter pits	+$222k
Sanitary	SDR-35, standard MHs	SDR-26 mains & leads, MH chimney seals, lot-line cleanouts, digital CCTV baseline	+$102k
Storm	12" @ 0.26% runs, no infiltration	500 LF upsized to 15"; 6 infiltration basins at sags	+$42k
MH electric service	105 × 100A + 39 × 200A	144 × 200A meter/main pedestals (NEC 550.32)	+$103k
RV power	Source not designed	200A+ service & feeder per NEC 551.73; metered pedestals	+$29k
Lighting	20 steel poles + 155 solar lamps	30 aluminum LED poles, #10 grounding conductor; solar lamps deleted	+$20k
Telecom	None	2" conduit along every road + stub to each site, 35 handholes, 24 spare road sleeves	+$164k
Buildings	Garage $110/SF (could be deferred); pavilion $75/SF	Insulated, heated shop with restroom & interceptor, built Phase 1; steel/metal-roof pavilion; utility stubs	+$99k
Site	Wood fence, paint, standard seed	Vinyl/composite fence, thermoplastic, prismatic signs, warrantied buffer & hydroseed	+$42k
QA / general conditions	9 months, $40k testing, 1.75% bonds	10 months (staged paving), $65k testing, 2.25% incl. 2-year warranty bond	+$64k + markups

Permitting note: Most changes are materials or specifications within the approved layout. Concrete curb and gutter, the added hydrants and loops, and the RV service will likely need an EOR revision letter to EGLE/LARA, but

should not reopen site-plan approval. Road and lot geometry are unchanged.

3. What I Deliberately Did Not Upgrade

Long-term thinking is not the same as gold-plating. These items stay as designed because the plan's choice is already the right long-term answer:

- **Asphalt roads, not concrete.** Concrete roads are roughly 2–2.5× the first cost, and a 4" asphalt section with mill-and-overlay is the lower-cost lifecycle answer at these traffic volumes.
- **HDPE storm, not RCP.** Dual-wall HDPE has a 75–100-year design life in sand, with no corrosion.
- **6" C900 water, not 8".** Upsize only if the hydraulic model or fire flow requires it. The supply is the existing 6" system, so upsizing downstream alone may buy little. The decision is flagged and should be made before pipe is ordered.
- **Utility-owned electric distribution.** Master-metering (VE A12) would put transformer and cable replacement on the owner forever.
- **No added amenities.** There are no rent-neutral amenities and no EV chargers. 200A services plus spare conduit make chargers a resident-level add later.

4. 40-Year Cost of Ownership

System	PV Rev 0	PV Rev 1	PV savings	Capex premium	Net benefit	Payback (yrs)
Pavement (3" → 4")	$318,177	$165,640	$152,537	$95,959	$56,577	25
Curb (HMA → concrete C&G)	$195,191	$47,946	$147,245	$196,475	($49,230)	53
MH service (100A → 200A)	$157,417	$0	$157,417	$102,930	$54,487	26
Site lighting (solar → poles)	$195,653	$17,264	$178,389	$19,834	$158,555	4
Privacy fence (wood → vinyl)	$38,941	$24,896	$14,045	$10,602	$3,443	30
Storm flat runs (12" → 15")	$18,822	$6,444	$12,377	$4,750	$7,627	15
Water dead ends (loops)	$31,668	$7,917	$23,751	$43,890	($20,139)	74
Sanitary MH seals	$38,861	$13,874	$24,987	$18,525	$6,462	30
Pavement markings	$58,214	$28,357	$29,857	$5,700	$24,157	8
TOTAL	**$1,052,945**	**$312,339**	**$740,606**	**$498,665**	**$241,940**	**27**

These are 2026 dollars at a 4% real discount rate over 40 years. Event costs and intervals are estimator judgment and editable on the Lifecycle tab. Two items show a net negative:

- **Curb and gutter.** This is justified by function, not present value. It gives a true gutter flowline, so inlets capture flow and the sags do not pond and ice (the C3 risk). It also takes plow strikes and home-delivery traffic, and it matches the specified catch-basin frames.
- **Water loops.** These are justified by fire flow, water quality, and not leaving dead ends that put homes out of water during repairs.

Not quantified, but strongly favoring Rev 1: conduit (about $9/LF now in open trench vs. $25–40/LF bored later under finished lots), 8" base (it cannot be added later), SDR-26 and lot-line cleanouts (fewer excavations), tracer wire, GIS as-builts, and owner-controlled driveways.

5. Capital Replacement Reserve

A permanent owner should fund replacements from NOI starting in Year 1. Based on the Rev 1 components (wearing course, crack seal, curb repair, castings, valves and hydrants, AMR meters, fixtures, RV pedestals, markings, fence, garage and pavilion), the straight-line reserve is about $33,500 per year, or $216 per site per year ($18/month) in today's dollars. Index it annually.

Underwriting often assumes $50–100 per site per year. For this asset that is too low, and it is the gap that leaves long-term owners with deferred maintenance. Underground mains and sewers (50–100-year life) belong in a long-range capital plan supported by periodic CCTV and condition assessment, not in the annual reserve.

6. Prior VE Memo Re-scored for a Permanent Hold

The 9/28 Cost & VE Memo optimized first cost and exit value: its "net value" was NOI capitalized at a 6.25% cap rate. For a forever owner, that lens is wrong for buried and structural items. The table below shows the re-scoring.

Item	Prior	Long-term verdict	Reason
A1 Two-way roads 24′ → 22′	Accept	REVERSE	Width is permanent: home replacements every 30–40 yrs, fire apparatus, parking, snow storage
A4 Valves 48 → ~30	Accept	REVERSE	Every main break for 50+ yrs is isolated by these valves
A5 Native sand backfill	Accept	MODIFY	Reuse is fine, but only with gradation and density testing at every lift
A6 Base 8″ → 6″	Accept	REVERSE	Base can never be added without reconstruction
A7 Cleanouts instead of MHs	Accept	REVERSE	MHs give permanent jetting/CCTV access
A8 Delete solar lamps	Accept	ADOPT	Replace with hard-wired poles
A9 Driveways to home package	Accept	MODIFY	Owner installs to one spec; recover cost in home price
A10 Defer garage & pavilion	Accept	REVERSE (garage)	Permanent ops base needed from day 1; pavilion may phase
A11 Walk → asphalt trail	Accept	REVERSE	Concrete 40+ yrs vs ~15–20
A2 / A12 Delete curb / master-meter	Reject	CONCUR	Curb upgraded to concrete C&G
A3 Infiltration basins	Pursue	ADOPT	Included
B1 +10 sites (narrower lots)	Accept	HOLD	Lot size is permanent; fit-test tomorrow's larger homes first
B4 Submetering / B6 Conduit	Accept	ADOPT (B6 mandatory)	Included
B5 SW lots → RV	Conditional	REJECT for long term	MH income is stickier; use only as a temporary absorption bridge

The full table, including the revenue-memo items, is on the "Prior VE – LT Review" tab. Net effect: the VE memo's $7.78M / $47.2k-per-site recommended plan should not be used as the budget. Its savings came mostly from cost transfers and from thinning buried specs.

7. Cost Breakdown by CSI Division

CSI Division	Rev 0	Rev 1	Δ	Rev 1 $/site
01 General Requirements	$831,694	$1,050,259	+$218,565	$6,776
02 Existing Conditions	$197,600	$197,600	$0	$1,275
03 Concrete	$422,397	$1,059,789	+$637,392	$6,837
06–09 Amenity Buildings	$195,776	$251,712	+$55,936	$1,624
22 Plumbing	$15,510	$41,830	+$26,320	$270
26 Electrical	$1,208,925	$1,497,107	+$288,182	$9,659
31 Earthwork	$477,324	$510,299	+$32,975	$3,292
32 Exterior Improvements	$797,843	$1,282,367	+$484,524	$8,273
33 Utilities	$2,356,789	$2,747,039	+$390,250	$17,723
TOTAL DIRECT HARD COST	**$6,503,858**	**$8,638,002**	**+$2,134,144**	**$55,729**

Soft-cost changes: grading plan + EOR redesign ($75k); GIS as-builts ($25k); geotech expanded to $40k; owner's rep raised to 3.5% for a full-time resident inspector. Soft total is $1,992,800 plus 10% contingency. Unit-cost basis and regional factors (0.92–0.95) are unchanged from Rev 0.

8. Contingency & Risk — Long-Term Responses

Contingency stays at 12% hard / 10% soft. One rule matters for a permanent owner: if bids come in high, re-phase or defer whole elements (for example, the pavilion). Do not buy back the budget by thinning buried or structural specs, such as base, pipe class, valves, conduit, curb or service size.

- **Wastewater (Moderate).** If the receiving system is private, its operation and eventual replacement are a permanent owner liability. Get a condition assessment, not just a capacity letter.

- **Electric (High).** Confirm Consumers Energy sizes transformers for 200A on every lot. Utility ownership keeps that equipment off the owner's books.
- **Fire flow (Moderate).** Run the hydraulic model now. Upsizing trunk mains during construction costs about $14/LF more; doing it later means replacement.
- **Construction quality (Moderate).** The plan review lists 9 critical issues (C1–C9). A forever owner inherits every defect, so Rev 1 funds the EOR fixes, hold-point testing, CCTV, a warranty bond and GIS as-builts.
- **Fees, geotech, stormwater.** Unchanged from Rev 0: $3,000/site fee allowance; geotech before pavement design; infiltration basins as a hedge.

9. Phase 1

With the same placeholders (60 sites, 60% of the linear network), Phase 1 is about $7.13M, or $118.8k per Phase 1 site. The cost is front-loaded because the maintenance shop, all tie-ins, the trunk utilities at full size, the fence, the buffer and the electrical racks land in Phase 1. Enter the real Phase I lot count from Sheet 06 on the Inputs tab.

10. Next Steps

- Issue owner direction to Spicer Group: long-term spec book (C&G, SDR-26, DR18 with tracer wire, 200A services, conduit and sleeves, MH seals, cleanouts, infiltration basins), plus resolution of plan-review items C1–C9.
- Hydraulic model and fire-authority letter to decide 6" vs 8" trunk and final hydrant count.
- Consumers Energy load letter for 200A on all MH lots and CIAC quote.
- Receiving sewer/water system: capacity, fee letters and condition assessment.
- Geotech for pavement design: confirm 8" (or 10" on two-way roads) base.
- **Excluded:** homes, setup and foundations, garages, land, gas distribution, telecom relocation by others, detention basin, off-site work, developer fee, escalation beyond 2026 Q3. MH driveways are now INCLUDED.
- Set the capital reserve policy ($216/site/yr, indexed) in the operating budget from Year 1.

Long-Term Hold -- Cost & VE Memo Rev 2 (full, 10/2/26) [long_term_hold_cost_ve]

MEMORANDUM – REVISION 2

The Bluffs on Manistee Lake – North Expansion: Cost & VE Re-tested for a Perpetual Hold

To: Marc Maricondo, Jennings Realty Inc.

Date: October 2, 2026

Supersedes: Section 5 and the recommended plan of the 9/28/26 Cost & VE Memo. The base estimate (Sections 1–4) is unchanged.

Companion: Bluffs_North_Expansion_Cost_Model_Rev2_LongTermHold.xlsx – new Lifecycle tab; VE and Rec Plan tabs rebuilt. Every number below comes from it.

Premise: The owner never sells. Every choice is judged by what it costs to OWN for 50+ years, not what it costs to BUILD or what it would sell for.

Bottom line

- **The metric changed.** The 9/28 memo ranked ideas by capital saved plus NOI capitalized at a 6.25% cap rate, which is a seller's yardstick. Rev. 2 ranks them by **50-year lifecycle NPV** in today's dollars at a 4.5% real discount rate: capital now, plus every future overlay, replacement, retrofit and repair, plus NOI over the hold.
- **Six of the nine cost cuts I recommended on 9/28 fail that test, and I'm withdrawing them:** narrower roads, a thinner road base, fewer water valves, cleanouts in place of manholes, an asphalt trail, and deferring the maintenance garage. Each saved $19k–$136k once and costs as much or more to own. Moving the driveways into the home price is set aside as a financing choice, not a saving.
- **Long-term-hold plan: 165 sites, ≈$9.60M, ≈$58.2k per site.** That's $1.10M more than the base as drawn ($8.51M, $54.9k/site) and $1.82M more than the 9/28 recommendation ($7.78M). In return: ≈$154k/yr more NOI (directional), ≈$640k of avoided future replacement and repair (present value), and a **50-year NPV of ≈+$2.9M** versus the base.
- **What the money buys:** a 4″ pavement with the wear course held until the homes are set; concrete curb; 200A service to every home; 15″ storm on the flat runs; SDR-26 deep sewer and tracer wire on the rear-lot water mains; fiber conduit in the open trench; recorded easements and GIS as-builts; a full-time owner's inspector; bluff-slope protection; and street trees.
- **Phase 1 (~60 sites) is now ≈$4.8M (≈$80k/site),** because it carries the trunk sewer, the water loops and the full-depth roads built once for full buildout. Plan for a ≈$330k clubhouse addition when Phase 2 fills: a 5,000 SF clubhouse can't serve ~320 households for decades.

__Still directional: MH rents have no comps. The RV seasonal rate now uses the subject__'s 2026 rate card ($6,995, top of the $4,830–6,995 seasonal range). Lifecycle timings are estimator judgment for northern-Michigan freeze-thaw, plow and MH-transport loading; the geotech pavement design should confirm them.

1. What a perpetual owner should judge differently

Question	Short-term / sale lens (9/28)	Perpetual-hold lens (Rev. 2)
How is an idea scored?	Capital saved + NOI ÷ cap rate	50-yr NPV: capital + PV of every future replacement/repair + NOI over the hold
Buried work	Meet code minimums	Built once: you can't cheaply reopen 155 back yards or new pavement
Pavement & curb	Thinnest section that passes	Section sized for 155 MH deliveries, garbage trucks and 50 winters of plowing
Electrical capacity	Size for today's homes	Size for electrification (heat pumps, EVs) – 200A everywhere
Lease-up speed	Faster cash (e.g., RV swap)	Stickier long-term residents. Absorption speed matters less than resident quality
Inspection & records	Contractor QC	Owner's inspector, CCTV, GIS as-builts – you inherit every defect
Trees, slopes, curb appeal	Cost with no near-term rent	Compounding assets; only a long-term owner captures canopy and bluff value

What already fits a perpetual hold and should NOT change: utility-owned electric distribution (no transformers to replace), gravity sewer with no lift station, C900 PVC water mains, the 8″ sewer sized well beyond buildout, and the existing amenity core.

2. Lifecycle tests (50 years, 4.5% real)

Base spec = as drawn. Positive net = the alternative is cheaper to own. Assumptions per item are on the Lifecycle tab.

Test	Capital Δ	PV future savings	50-yr net	Breakeven	Result
LC1 Pavement 3″ → 4″ HMA, wear course after home set	+$97k	+$217k	+$119k	~22 yrs	Upgrade
LC2 Asphalt curb → concrete curb	+$171k	+$154k	–$17k	~50 yrs	Near break-even – upgrade (see VE L2)

Test	Capital Δ	PV future savings	50-yr net	Breakeven	Result
LC3 100A → 200A on single-wide sites	+$51k	+$122k	+$71k	~21 yrs	Upgrade
LC5 12″ storm @0.26% → 15″	+$4k	+$24k	+$20k	~8 yrs	Upgrade
LC6 SDR-26 deep sewer + water tracer wire	+$11k	+$20k	+$10k	~26 yrs	Upgrade
LC4 Delete per-lot solar lamps (prior A8)	−$80k	+$102k	+$183k	—	Cut stands
LC7 Road base 8″ → 6″ (prior A6)	−$44k	−$83k	−$38k	—	Cut withdrawn
LC8 Concrete walk → asphalt (prior A11)	−$22k	−$17k	+$5k	—	Wash; keep concrete
LC9 Defer maintenance garage (prior A10)	−$136k	−$175k	−$39k	—	Cut withdrawn
LC10 Terminal cleanouts vs. MHs (prior A7)	−$19k	−$21k	−$2k	—	Cut withdrawn

3. Every alternative re-tested

Alternative	9/28 verdict	Rev. 2 verdict	50-yr NPV	Why
A1 Two-way road 24′→22′	Accept if >22′	REJECT	≈$0	Edge failure, snow storage, fire access (error reg. C9/P11)
A4 Valves 48 → 30	Accept	REJECT	+$55k	Each break would shut off twice as many homes, forever. Reliability can't be added later
A5 Native sand backfill	Accept	ACCEPT + testing	+$29k	No durability penalty if gradation/compaction verified
A6 Base 8″ → 6″	Accept	REJECT	−$41k	LC7
A7 Cleanouts for 4 MHs	Accept	REJECT	−$4k	LC10: jetting/CCTV access for 50+ yrs
A8 Delete solar lamps	Accept	ACCEPT	+$182k	LC4: also lifecycle-positive
A9 Driveways in home price	Accept	NEUTRAL	n/a	Financing choice; out of totals. Community pours to its spec and owns them
A10 Defer garage	Accept	REJECT	−$39k	LC9: a ~320-site operation needs a shop day one
A11 Asphalt trail	Accept	REJECT	+$3k	Dollar wash; concrete is better for a 55+ community
A2 / A12 Curb deletion / master-meter	Reject	REJECT	−$95k / −$712k	Even clearer for a forever owner
B1 +10 sites	Accept	ACCEPT w/ floors	+$628k	Perpetual NOI. Keep lots ≥6,000 SF, 20′ separation, garages, nothing on the bluff edge
B2 RV/boat storage	Accept	ACCEPT	+$396k	Build with a paved apron and a real gate
B3 Premium pricing	Accept	ACCEPT	+$225k	No capital
B4 Water/sewer submetering	Accept	ACCEPT	+$1.18M	Caps decades of utility-rate exposure; meter reserve included. Only works if the community pays a master bill
B5 SW → RV swap	Consider	REJECT	−$284k	Homeowners are stickier than seasonal RV guests
B6 Fiber conduit	Accept	MUST-DO	+$141k	Trench is open once
B7 Clubhouse addition	Defer	Phase 2 commitment	−$271k	Negative on direct NOI; it's for retention of ~320 households
C2 Water loops	Accept	ACCEPT	−$8k	Water age, fire flow, flushing

New durability items (L-series), all in the plan: L1 pavement (+$116k NPV), L2 concrete curb (−$25k, near break-even), L3 200A everywhere (+$71k), L4 storm upsizing (+$18k), L5 pipe durability (+$10k), L7 recorded easements + GIS as-builts (+$5k), L8 full-time owner's inspector + CCTV acceptance (≈$0), L9 Kentucky hydrant + fire-flow test (−$12k), L10 bluff-slope protection (−$15k), L11 street trees (−$61k, strategic). The last four are insurance and community quality. A forever owner buys them because the downside they prevent is permanent.

4. Recommended long-term-hold plan

The five moves	Hard Δ	Soft Δ	NOI Δ/yr	50-yr NPV
1. Build it once – durability (L1, L2, L4, L5, L9, L10)	+$325k	+$18k	—	+$92k
2. Future-proof utilities & operations (L3, B6, C2, L7, L8)	+$98k	+$102k	+$8k	+$206k
3. Revenue that lasts (B1 +10 sites, B2 storage, B3 premiums)	+$235k	+$57k	+$79k	+$1.25M
4. Utility cost pass-through (B4 submetering)	+$70k	+$5k	+$66k	+$1.18M
5. Only the cuts that are also lifecycle-positive (A5, A8)	−$113k	+$4k	—	+$211k
Phasing with build-once rules (C1) + street canopy (L11)	+$32k	+$25k	—	−$86k
Package (direct, before GC/contingency gross-up)	**+$646k**	**+$211k**	**+$154k**	**+$2.86M**

Full buildout	Base (as drawn)	Long-term-hold plan	Δ
Sites	155	165	+10
Total hard cost (incl. GC, bonds, 12% contingency)	$6.95M	$7.73M	+$0.78M
Total soft cost	$1.56M	$1.87M	+$0.32M
TOTAL PROJECT COST	**$8.51M**	**$9.60M**	**+$1.10M**
Cost per site	$54,900	$58,200	+$3,300

Full buildout	Base (as drawn)	Long-term-hold plan	Δ
Stabilized NOI (directional)	$623k	$777k	+$154k
Cash yield on cost	7.3%	8.1%	+0.8 pts
PV of avoided future replacement/repair	—	$640k	
50-yr NPV vs. base (real $)	—	+$2.86M	
Phase 1 (~60 sites)	—	≈$4.80M (≈$80k/site)	
Planned Phase 2 add: clubhouse expansion (outside totals)	—	≈$330k	

Why this costs more than the 9/28 plan ($7.78M): the driveway cost shift (≈$0.85M grossed up) is out of the totals, four cuts that were in the 9/28 totals (valves, road base, cleanouts, trail) are reversed, and ≈$0.4M of durability and future-proofing work plus ≈$0.1M of owner inspection, easements and as-builts is added. The +10 sites, storage yard and submetering are carried in both plans. Ownership cost per site over 50 years is lower, not higher.

5. Build-once rules for the contract

- Two-year contractor warranty on all buried utilities and paving (not one), backed by a maintenance bond; 11-month warranty walk with CCTV re-run on any suspect sewer/storm run.
- CCTV and mandrel acceptance of every sewer and storm run, pressure and bacteriological tests, and tracer-wire continuity, all before any HMA. No paving over unaccepted pipe.
- Leveling course only until each phase's homes are delivered and set; wear course after. Casting final-set after paving (error register C3).
- Trunk mains, capacity reservations and road sections built for full buildout in Phase 1. No temporary pavement, no undersized interim mains, no permanent dead ends.
- Recorded utility easements and no-build zones on every rear-lot main (sheds and garages kept off); valve boxes and curb stops in accessible, mapped locations; survey-grade GIS as-builts delivered as a condition of final payment.
- Attic stock: spare castings, valve-box parts, meter pedestals, light fixtures and curb-stop parts matched to the installed brands.
- Resolve the error register's critical items (grading plan, storm-inlet geometry on the 16′ roads, water/sewer separations, RV power source) before mobilization. Those are design defects you'd otherwise own forever.

6. Long-term risks to add to Section 4

Risk	Rating	Long-term view
Ownership of water/sewer beyond the connection points	HIGH	If the park's water/sewer is private (well, lagoon or plant), you own its replacement and regulatory compliance forever. Get the system's age, condition and capacity before committing 155 more ERUs, and reserve capacity for full buildout.
Rear-lot water mains	MODERATE	Fine to own if easements are recorded, structures are kept off, and the mains can be located (tracer wire + GIS). Otherwise every repair is a back-yard dig through a resident's shed or garden.
Pavement fatigue on a thin section	MODERATE	MH transports and garbage trucks fail thin sections early. Mitigated by L1 and the delayed wear course.
Bluff erosion (67F soils)	MODERATE	Irreversible and the community's signature view. Mitigated by L10 and a no-mow buffer.
Electrification load	MODERATE	Mitigated by L3 (200A). Confirm the utility sizes transformers for it.

Next steps

- Decide which L-items to issue as addenda before bid. L1, L2 and L4 change the drawings and fold into the storm-inlet redesign the error register already requires.
- Ask Spicer for a geotech-based pavement design (traffic: MH transports + refuse trucks, 50-yr horizon) to confirm the 4″ section.

- Confirm the receiving water/sewer system's ownership, condition and capacity for full buildout. If private, add a 50-yr capital plan for it.
- Confirm with the electric utility that 200A on all 144 MH sites is acceptable and transformers are sized for it.
- Provide MH rent comps so the B-series NOI figures can be underwritten rather than directional.

Bluffs North Expansion – Rev. 2 Perpetual-Hold Memo | Class 3 estimate, directional NOI | Page

Long-Term Hold -- Revenue & VE Within Approved Plans (full, 10/2/26) [long_term_hold_revenue]

Bluffs North – Revenue & VE Within Approved Plans (rev. 10/2/26, permanent-hold basis)

Doc: https://claude.ai/code/artifact/b5852eca-a852-419e-9e7e-fd75545eed20 · Model: Bluffs_North_Amenity_ROI_Model.xlsx

- **Owner directive (10/2):** permanent hold, never selling. Judge every item on lifecycle cost; no short-term corner cutting.
- **Method:** NOI after replacement reserve (capex × replacement share ÷ life). Shows 30-yr NPV and perpetual NPV at 7% discount and 3% growth. Cap-rate value is for refinancing reference only.
- **Totals (recommended, long-life specs):** $77k/yr NOI after reserves on $397k, a 19.4% long-term yield. 30-yr NPV $0.91M; perpetual $1.53M. Without the water pass-through: $41k on $275k (NPV $0.42M / $0.74M).
- **Top items:** Tier A/B pricing; metered RV pedestals (89% yield, tie to C7); water/sewer pass-through (counsel; meter settings in the M5 detail); super-site screening.
- **Super-site package** (screening, stamped patio, shed, fire table) at long-life spec: ~$10.2k/site, 5.6% yield; ~flat over 30 yrs, +$45k in perpetuity. Fire table is the drag.
- **Reversed:** PT-wood fence VE withdrawn. Vinyl is ~$58k vs ~$93k over 30 yrs. Pavilion: durable galvanized kit with standing-seam roof, ~$92k vs $129k site-built timber over 30 yrs.
- **Dropped after reserves:** golf carts, MH shed lease, EV Level 2. Fitness room is optional (retention only).
- **Added:** a shade tree on every lot (negative over 30 yrs, +$35k perpetual). Durability specs to lock in: P14 pipe classes, #10 EGC, castings, 4,000 psi air-entrained concrete, pavement preservation, solar-lamp battery reserve, keep-out zones over mains, GIS as-builts.
- Subject 2026 RV rates: super site $1,925/mo, deluxe $1,325, seasonal $4,830–6,995.
- Excluded as plan amendments: +10 sites (B1), RV/boat storage (B2), courts, playground, lake access, entry sign, MH patios, new conduit.

Long-Term Hold Addendum -- Register Changes & Next Steps (10/2/26) [long_term_hold_addendum]

Bluffs North Expansion – Long-Term Hold Addendum (10/2/26)

Owner intent (Marc, 10/2/26): We hold this community forever and never sell. Every decision is judged on 40–50-year ownership cost, not first cost. **Supersedes:** the VE recommendations in Bluffs_North_Expansion_Cost_VE_Memo.docx, the VE items in the revenue/VE doc, and the field corrections in the plan-review register wherever they conflict with this addendum. **Model:**

Bluffs_North_Expansion_Class4_Estimate_LongTermHold.xlsx. It is the Class 4 workbook with a new "Long-Term Hold" tab and a Summary block. Every line has a 1/0 toggle and live formulas.

Revised numbers (Class 4 basis)

	Class 4 base	Long-term hold	Δ
Total project cost	$8.97M	**$10.51M**	+$1.53M (+17%)
Cost per site	$57.9k	**$67.8k**	+$9.9k
Driveways built by the park, priced into homes (recoverable)	excluded	+$0.79M	recovered in home sales
Avoided future cost, 40–50 yrs (directional)	—	≈$1.9M	≈2.4× the discretionary spend

- About $0.42M of the +$1.53M is not discretionary. It corrects the base estimate: the two-way roads were priced at a 14′ mat (+$163k direct, pending C9), plus the WM re-profiles, the RV service, the Montana inlet and frost-protected site risers.
- About $0.94M is durability upgrades with Div 01, contingency and soft costs added. The remaining ~$0.18M is soft line items: a resident inspector, GIS as-builts, redesign, geotech, warranty and a reserve study.
- The Phase 1 tab doesn't include these adjustments yet. Most of them are linear, so roughly 60% lands in Phase 1.

VE decisions reversed or kept

Item	Before	Now	Why
A1 two-way road 24′→22′	Accept	**Reject**	Width is permanent; it serves plows, parking and fire
A4 valves 48→30	Accept	**Reject** (keep base valving)	Isolate a break without shutting down a block
A6 road base 8″→6″	Accept after geotech	**Reject**; go thicker (LT2)	50-yr pavement life
A7 terminal cleanouts in lieu of MHs	Accept	**Reject**	Jetting/CCTV access for life
A9 driveways to home buyer/dealer	Accept (transfer)	**Reverse**: park builds to its own spec, price into home (RC1)	Park owns the land and the drive forever
A10 defer garage	Accept	**Reverse**: build in Phase 1, upgraded (LT20)	1.2 more miles to plow; equipment indoors
A11 concrete walk→asphalt	Accept	**Reject**	Concrete lasts 50 yrs; 55+ walkability
B1 +10 sites via narrower lots	Accept	**Drop**	Smaller lots hurt retention and resale of homes for decades; also reopens permits
Pavilion light-duty kit	Accept	**Reject**; durable frame (LT21)	50-yr structure
Fence material VE	If allowed	**Upgrade** to vinyl/composite (LT22)	No wood replacement cycles
Wi-Fi rough-in only	Conductor rough-in	**Conduit to every site** (LT6)	Avoids boring finished lots
Tree on every lot	Not recommended	**Do it**, front yard, utility-safe species (LT23)	Canopy pays back over decades
A8 delete solar lamps	Accept	**Keep** (LT7), add poles if photometrics need them (LT8)	Removes a perpetual battery-replacement program
A5 native sand backfill	After tests	**Keep**, only with gradation tests passing MDOT Class II	Sound engineering, not a corner cut
A3 infiltration hedge	Pursue	**Keep**, with sump hoods/pretreatment (LT16)	Unprotected infiltration clogs
A2 curb deletion, A12 master-meter	Reject	**Reject**	Unchanged
B4 water/sewer submetering	Accept if master bill	**Keep**; specify 20-yr radio AMR in frost-proof pits	Unchanged
B2 RV/boat storage	Accept	**Later plan amendment only**	Unchanged

Durability upgrades (Long-Term Hold tab, section A)

LT1 concrete mountable curb & gutter ($196k). LT2 4″ HMA ($86k). LT3 wearing course delayed until homes are set ($24k). LT4 one-way loops at 20′ BOC ($55k, subject to permit). LT5 200A at every MH site ($69k). LT6 comms conduit ($58k). LT7/LT8 solar lamps out, poles in (–$68k net). LT9 loop all dead ends ($27k). LT10 tracer wire ($16k). LT11 8″ backbone, only if fire flow requires it (off). LT12 SDR 26 sewer ($19k). LT13 chimney seals ($13k). LT14 warranty CCTV ($22k). LT15 15″ storm in place of 12″ at 0.26% ($6k). LT16 sump hoods ($13k).

LT17 HDPE rings and heavy castings ($17k). LT18 #10 EGC ($2k). LT19 200A panel G ($4k). LT20 garage upgrade ($51k). LT21 pavilion ($13k). LT22 fence ($22k). LT23 street trees ($62k).

Changes to the plan-review register

- **C2:** Before relocating the storm structures to fit a 16′ road, ask Spicer for the base-bid one-way width and price it (LT4). A 20–21′ road matches the storm layout as drawn, gives snow storage, and lets a parked car leave fire and home-delivery access open. If the 16′ alternate stays, relocate as written.
- **C3/M4:** Build concrete curb & gutter (LT1). This makes the EJIW 7045 frame correct and sets inlets in a concrete gutter with a defined flowline.
- **C5/C6:** Re-profile only. Never accept a reduced-separation variance.
- **C7:** Size the RV service for full 551.73 demand plus one spare pedestal position.
- **M3:** Upsize to 15″. This is no longer "or steepen".
- **M5:** Each site detail must include a curb stop and box, a frost-protected water riser/meter pit, a 4–6″ sewer riser with cleanout, and the comms conduit end.
- **M6:** 200A meter/main pedestals at every MH site.
- **M7:** The garage is in Phase 1 scope, with stubs for water, sewer, 200A power and comms.
- **P9:** Loop every dead end. 8″ backbone if fire flow requires it.
- **P11:** Request a width alternate and No-Parking signage regardless of the outcome.
- **P14:** SDR 26 throughout, C900 DR18, tracer wire with test stations.
- **New hold points:** CCTV of every sanitary and storm main at acceptance and again at month 11 before warranty release. Mandrel test HDPE storm. Resident inspector on site for all underground work. GPS every valve, curb stop, wye and conduit end before backfill.

Ask Spicer / owner next

1. The base-bid road widths for one-way and two-way roads, and which widths the MHC construction permit was issued on.
2. A redesign package for LT1, LT2, LT4 and LT9 plus the register fixes, as one set before bid award.
3. Geotech scope for a 50-yr pavement design and bluff stability.
4. Consumers Energy: transformer sizing for 200A at every MH site.
5. Water/sewer ownership. If the system is private, the long-term operating and treatment liability belongs in the hold underwriting.
6. Commission a capital reserve study at stabilization.

FILES

Cost & VE Memo (full) ↗ | Class 4 Estimate Report (full) ↗ | Cost Model (live workbook) ↗ | Class 4 Estimate (live workbook) ↗ | Amenity ROI Model (live workbook) ↗ | Full 49-sheet plan set (page images + text) ↗ | Plan index (Sheet 1 of 49) ↗ | Phase 1/2 Bid Quantities -- 23ft BOC Base Bid (Addendum No. 6, Oct 2 2026) ↗ | Phase 1/2 Bid Quantities -- 16ft BOC Alternate No. 1 (Addendum No. 6, Oct 2 2026) ↗

Long-Term Hold -- Class 4 Estimate Rev 1 (live workbook) ↗ | Long-Term Hold -- Class 4 Estimate Rev 1 (report) ↗

Long-Term Hold -- Cost Model Rev 2 (live workbook) ↗ | Long-Term Hold -- Cost & VE Memo Rev 2 (full) ↗

Long-Term Hold -- Amenity ROI Model (live workbook, Lifecycle tab) ↗

CROSS-RUN NOTES (LOG)

1. The full Cost & VE Memo and the full Class 4 Estimate Report (read in full 2026-09-28, not just their summaries) both independently identify the same two open items as the biggest real cost risk: (1) whether the existing sewer/water system has capacity for 155 new connections, and (2) what Consumers Energy will charge for the electric line extension (CIAC) -- neither is on the drawings and each could move the total by $0.5-1.2M.
2. RV rents differ across documents: the Cost & VE memo assumes $3,500/mo seasonal RV (a dated public profile); the Ideas & ROI doc uses the subject's actual 2026 rate sheet ($1,925/mo super, $1,325 deluxe, $4,830-$6,995/season) and says these supersede the older figure. Trust the Ideas doc's rents.
3. The Class 4 estimate ($8.97M) and the Cost & VE Memo's base case ($8.51M) differ by about $466k (5.2% of the lower figure) on the same plan set. Both memos flag the same missing information (grading plan, geotech, tap fees) as the reason neither should be treated as final.
4. The Cost & VE Memo goes a real step further than the Class 4 report: it recommends an alternative 165-site plan (+10 sites) at $7.78M ($47,200/site), a $0.72M reduction from its own $8.51M base case, driven mainly by shifting driveway/shed cost into the home-sale package and adding water/sewer submetering revenue.
5. Both memos independently flag the same real electrical-tag discrepancy (fewer E1/E2/E3 tags legible than the 155 real sites) but neither should be read for an exact count -- see the plan-index verification note below.
6. Added 2026-09-28: an independent second review (model claude-sonnet-4-6) was run against the raw 49-sheet plan text alone, with zero exposure to the outside consultant's (Marc's) memos or register -- see 'independent_review_raw' and the per-question 'Independent analysis' toggles under The big picture above.
7. Construction status update, 2026-10-02 (Andrew Chime, Grex Dev -- the project's construction manager -- weekly email to Jay/Marc): the Phase 1 Bid Package (Site Development) and Phase 2 Bid Package (Future Homesites) went out as Addendum No. 6 on Oct 1, 2026. Contractor bids are due Tuesday Oct 6 at noon ET (Walton confirmed receipt; AJ's, Tridonn, and Leopardo also bidding -- Leopardo needs until ~Oct 20 for a full bid). Grex plans to deliver a bid comparison Wednesday Oct 7 and review it with Jay/Marc Thursday Oct 8.
8. Real quantity revisions since the AI cost estimate was produced, all in the Phase 1 Bid Package (per the project's own RFI Log, RFI 006/014/016, consolidated at RFI 031, confirmed directly from the real Oct 2, 2026 bid-package quantity sheets): Water services cut from 7,800 LF to 3,120 LF; 6" sanitary leads cut from 9,890 LF to 3,432 LF; electric panelboards increased from 1 to 2; and site-service/RV pedestals added (81 at 100A single-phase, 63 at 200A single-phase, 11 RV loop-fed pedestals). These are large swings on two of the cost estimate's bigger unit-price lines -- treat the AI Cost & VE Memo's water-service and sanitary-lead dollar figures as pre-revision and likely overstated until the real contractor bid comparison (expected Oct 7) supersedes them. The two real MDOT-format quantity sheets are saved and linked below.
9. Jay's own cost idea, 2026-10-02, now in front of Grex: whether Consumers Energy and DTE (the two utilities trenching their own lines today) would allow a single contractor to trench for all three utilities in one pass instead of three separate trenching mobilizations -- Jay's words: "That should save significant money just having one sub do the trenching than 3 subs doing trenching at different times." Grex is asking both utilities whether they'll allow a joint trench or credit Jennings for self-performing it; the GC bids already carry a per-foot price for trenching the primary (MDOT items U-47/U-48) so a real cost comparison will exist once the utilities answer. This is a genuine construction-sequencing savings, not a durability tradeoff -- it does not conflict with the long-term-ownership framing added the same day (see below).

10. A real schedule risk worth tracking: DTE's own builder guidance sets an October 1 deadline for utility applications/payment in counties with winter road restrictions (Antrim, Charlevoix, Cheboygan, Emmet, Leelanau, Otsego, Roscommon, Traverse City, the U.P.) -- Manistee County (where the Bluffs sits) is not on that list, but DTE itself says the list isn't exhaustive. Grex is asking DTE directly whether the cutoff applies here and whether it affects installation timing; unresolved as of Oct 2.

11. Permit status, 2026-10-02: LARA (Michigan's state construction-permit authority) review in progress; the sanitary sewer permit is under review by the local reviewing engineer (Aaron Cline); the water main permit review starts once LARA approves -- a real sequencing dependency, not yet flagged elsewhere in this review.

12. Added 2026-10-02, per Jay's direct instruction after reviewing the AI items: Jennings is a permanent, long-term owner-operator of every community it builds, never a short-term flipper. Jay's words: "we should add a prompt telling it that we are long term owners... That should change a bunch of the short term items it suggest to cut costs... you can cut so many corners and save cash short term, but they do come at a cost to haunt you in the long term." This is now a standing instruction in every stage of the AI plan-review prompt pack (see data/development_prompts/v1/_system.md) -- it will apply to any FUTURE review run on this or any other project, but does not retroactively change the cost/value-engineering findings already published below; re-running this review's cost/VE/ideas stages against the current prompt pack would be the way to get those findings reassessed through this lens.

13. MAJOR UPDATE, 2026-10-02: the outside consultant (Marc's own Claude project, independently of this dashboard) re-ran the full Cost Model, Class 4 Estimate, and Revenue/Amenity ROI analysis for a permanent, never-selling ownership basis -- the same long-term-ownership instruction Jay gave this project the same day (see the cross_run_notes above and data/jay_operating_principles.json's 'long_term_owner_never_flipping' entry), arrived at independently by Marc on his side. Four new full documents are added above: the Class 4 Estimate Rev 1 report, the Cost & VE Memo Rev 2, the Revenue & VE analysis on a permanent-hold basis, and a register-changes addendum. The ORIGINAL (9/28) base-case documents and numbers are kept, not replaced -- this is additive, same as every other revision on this page.

14. Two new long-term-hold estimates exist and they disagree by about $2.27M, for a real, explainable reason (same pattern as the original Class4-vs-CostVE disagreement above): the Class 4 Estimate Rev 1 ($11.87M, 155 sites, $76,559/site) re-estimates the AS-DRAWN plan with every durability/long-term spec applied uniformly, with no revenue-side offsets. The Cost & VE Memo Rev 2's recommended plan ($9.60M, 165 sites, $58,200/site) is a hand-picked plan that keeps the SAME durability upgrades but also keeps several revenue-positive ideas the long-term lens still supports (the +10 sites, RV/boat storage, water/sewer submetering) -- its own text states plainly 'ownership cost per site over 50 years is lower, not higher' than the base case. Read: Rev 1 answers 'what does the plan as drawn cost to build right'; Rev 2 answers 'what should Jennings actually build.' Rev 2's $9.60M recommended plan is the more actionable number for budgeting purposes.

15. What reversed, in both documents, independently: six of the original nine cost-cutting VE ideas from the 9/28 memo are now REJECTED once scored on a 40-50 year lifecycle basis instead of a seller's cap-rate lens -- narrower two-way roads, a thinner 6" road base, fewer water valves, cleanouts in place of manholes, an asphalt walking trail, and deferring the maintenance garage. Each saved $19k-$136k once and was found to cost as much or more to own over the life of the asset -- directly the pattern Jay described in his own email ('you can cut so many corners and save cash short term, but they do come at a cost to haunt you in the long term').

16. What's new, not just reversed: 200A electrical service on every manufactured-home site (not just doublewides), concrete curb and gutter instead of asphalt, 4" pavement with the wearing course delayed until homes are set, SDR-26 sewer with tracer wire, fiber/telecom conduit installed in the open trench now (quoted at $9/LF now vs. $25-40/LF bored later under finished lots -- a concrete, well-reasoned long-term-vs-short-term cost comparison), a capital replacement reserve policy ($216/site/yr, not previously addressed at all), and a full-time owner's inspector with CCTV/GPS as-built documentation requirements.

17. Register-item guidance updated (not the original findings -- those stay as originally reported for the historical record, see the Plan Issues register above): items C2, C3/M4, C5/C6, C7, M3, M5, M6, M7, P9, P11, and P14 now have revised field-correction guidance reflecting the long-term-hold lens (e.g. C3/M4 should get concrete curb & gutter, not just a re-profile; M3's storm upsize to 15" is 'no longer or steepen' -- now mandatory). See the new Long-Term Hold Addendum document above for the item-by-item detail.

18. Revenue side (Amenity ROI, permanent-hold basis): recommended long-life amenity package now shows $77k/yr NOI after replacement reserves on $397k of amenity capital -- a 19.4% long-term yield, 30-year NPV $0.91M, perpetual NPV $1.53M. The water/sewer submetering pass-through is the single biggest driver of that NOI. A pressure-treated-wood fence VE idea was withdrawn in favor of vinyl once a 30-year replacement-cost comparison was run (~$58k vinyl vs. ~$93k wood over 30 years) -- the same build-once logic as the cost side.

19. All 5 new source documents (3 xlsx, 2 docx) are uploaded and linked above -- including the Amenity ROI model's long-term-hold version, which downloaded under a misleading filename ("_1", looking like a plain duplicate) but is actually a distinct workbook with a Lifecycle tab in place of the original's VE tab; confirmed by comparing sheet names, not just the filename. Still open: reconciling the two disagreeing recommended-plan numbers ($11.87M vs $9.60M) with Jay/Marc before either becomes the real working budget. The original plan set PDF (03A Plan Set, REV 3) was also re-downloaded 2026-10-02 but is unchanged -- still REV 3, the same 49-sheet set already reviewed, not a new revision.

20. Added 2026-10-05: a structured placeholder (see 'bid_comparison' above/below) is now ready to capture Grex Dev's real contractor bid total the moment it lands (~Oct 7) against all four of this page's published cost estimates. This is the first real, external check this whole AI plan-review pipeline has ever had against actual money -- update it as soon as the real number is known, don't let the window pass.

21. Zoning check, 2026-10-05: Manistee Township's own 212-page ordinance has a High-Risk Erosion Overlay (55ft/35ft bluff setback) -- but that state program covers Great Lakes shorelines only, not inland Manistee Lake. Likely doesn't apply here; confirm with the township to close it out. Standard height limit townshipwide: 35ft / 2.5 stories.

22. Cross-model check (Gemini), 2026-10-05: independent review, zero exposure to this page's findings. New: 6" water mains feeding hydrants directly may be undersized for fire flow (NFPA 1) -- get Fire Marshal sign-off. One-way roads at 14ft are under the ~20ft fire-code minimum. Full text under each question's "cross-model second opinion" toggle.

23. Insurance cross-check, 2026-10-06: this page's own budget already assumes ~0.35% of hard cost for builder's risk/bonds/insurance -- but the company's separate insurance-schedule project (Ricky's audit, Sep 2026) found the ACTUAL policies may not be adequate (see action items above). Budgeted for, not yet confirmed secured. Source: ricky insurance project's compliance_report.json + HANDOFF.md -- not independently re-verified here, insurance project is a separate effort and was not touched.